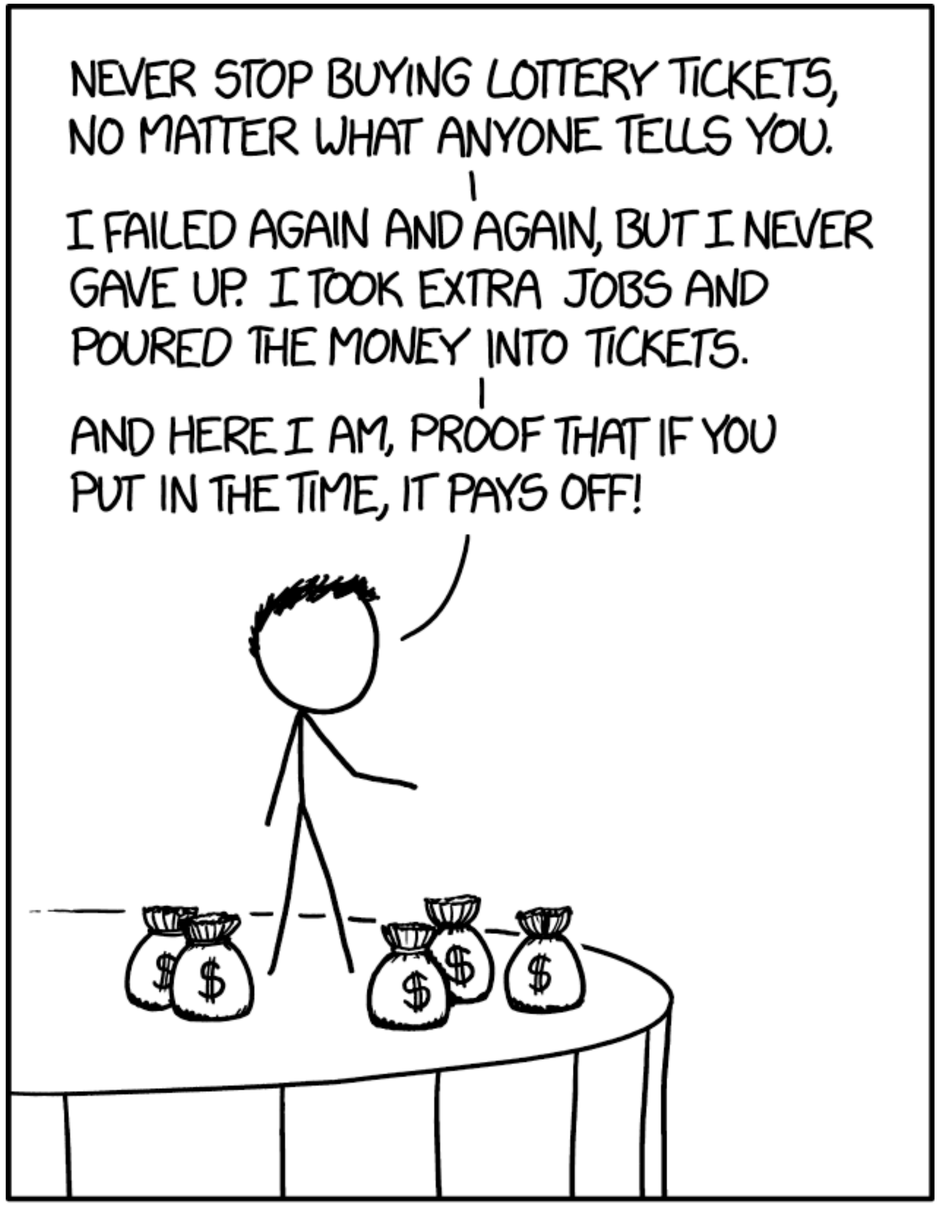
From paper to publication: Three things I wish I learned sooner

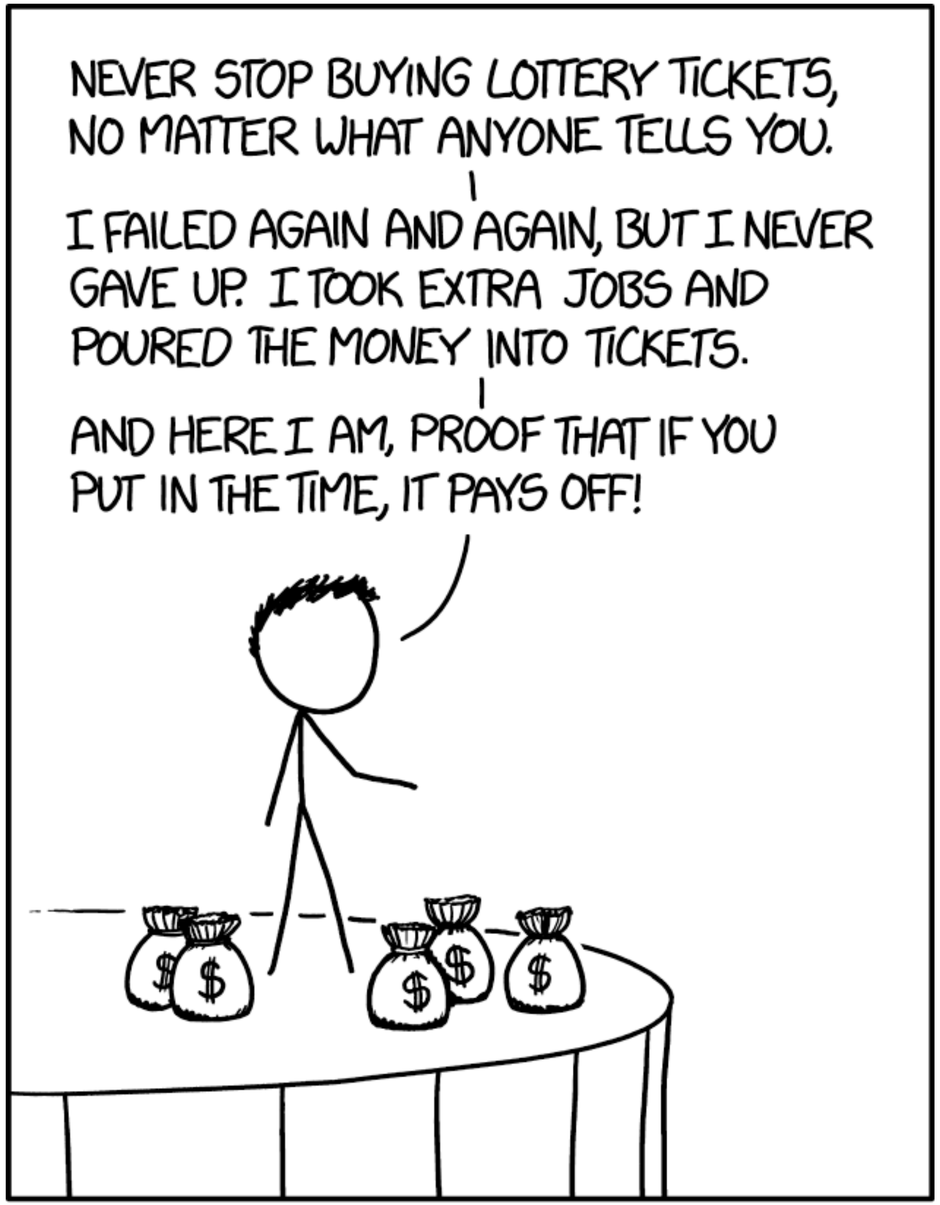
Mathew Kiang, ScD
Assistant Professor
Epidemiology and Population Health
Stanford University
mkiang@stanford.edu

Stanford MEDICINE | Epidemiology and Population Health



EVERY INSPIRATIONAL SPEECH BY SOMEONE SUCCESSFUL SHOULD HAVE TO START WITH A DISCLAIMER ABOUT SURVIVORSHIP BIAS.

https://xkcd.com/1827/



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Disclaimers

- Publishing is really random and I've been really lucky
- Based on my roles as
  - a public health researcher
  - section editor at a digital health / bioinformatics journal
  - associate editor at a clinical / specialty journal
- **Some of this won't apply to other fields!**

https://xkcd.com/1827/

"Assume a spherical cow..."

Initial submission

- You pick a journal
- You format your paper to the journals specs
- You write a cover letter

Technical review

- Check for journal format, completeness, coauthor contact information, etc.
- Initial screening for ethics and plagiarism
- Potentially send back to authors to fix issues

Senior editor triage

- Assess paper fit, scope, and priority
- "Is this paper likely to make an impact?"

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For selective journals,
50-80% of papers get rejected here.

See: `github.com/mkiang/paper_to_publication` for data/sources.

Associate editor review

- Assess paper for fatal flaws and novelty
- At some journals, AEs can also desk reject
- Finds 2 and 6 reviewers (often after inviting 20+)

Peer reviewers

- Diverse perspectives: experts in this subfield to researchers in adjacent fields

AE synthesizes

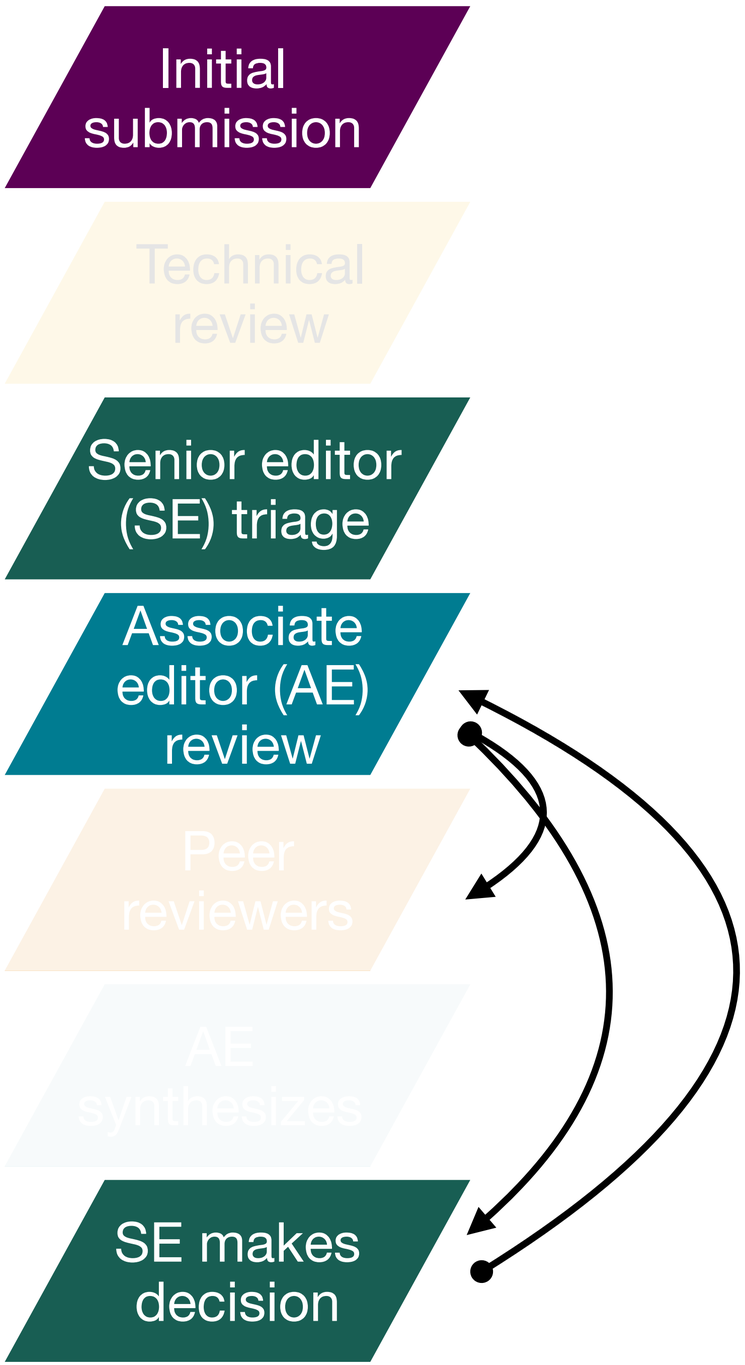
- Reads all reviews, re-reads paper, synthesizes (potentially conflicting) peer reviews
- Often adds their own review
- Writes a recommendation to senior editor (or editorial board)

SE makes decision

- Makes a final decision
  - Accept
  - Revise and resubmit (R&R)
  - Reject (with or without transfer)

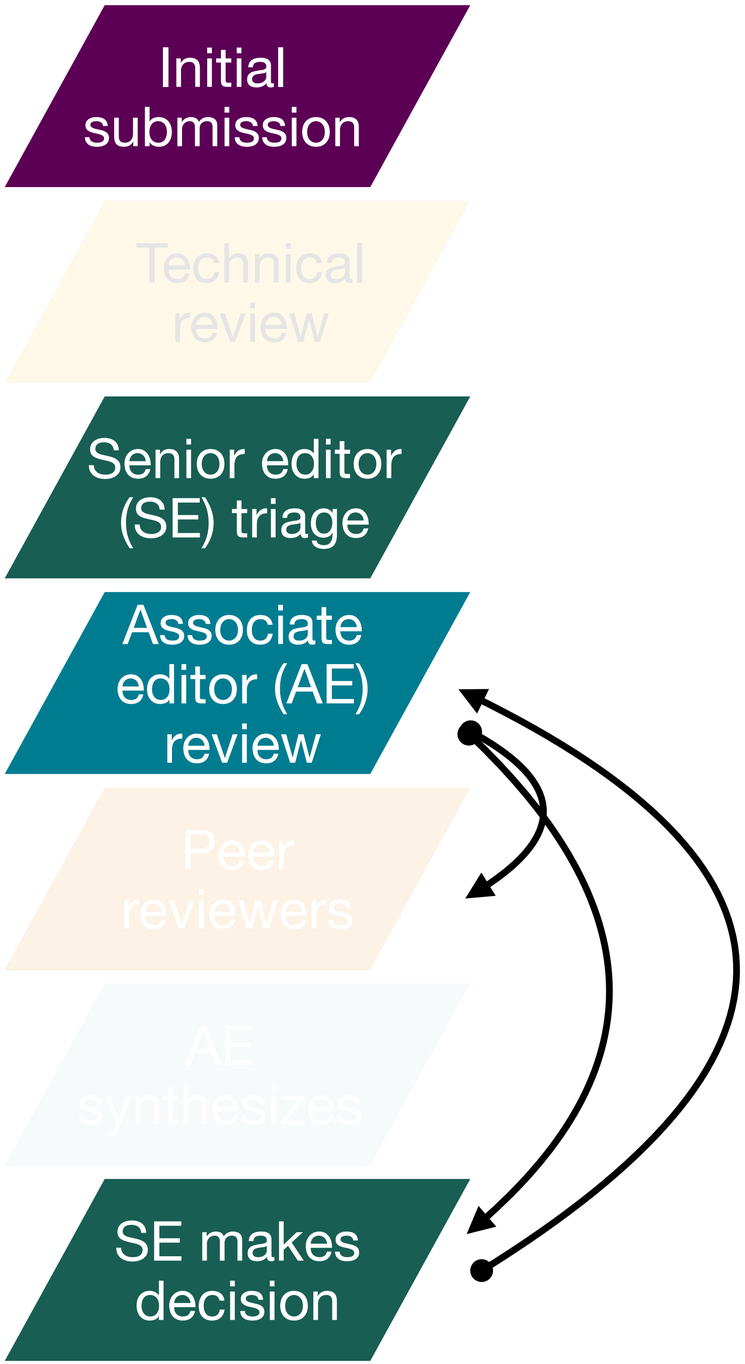
R&Rs

- Authors provide revisions
- Goes back to associate editors
- AE decide if it needs to go back to reviewers again and process repeats



Focus

1. Initial submissions
2. Senior editor triage
3. Revise and resubmits



Focus

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Not talking about

1. Importance of publications
2. Doing good science or reproducibility
3. Cost
4. Quality vs quantity
5. h-index or journal impact factors

Agenda

1. **Embrace rejection**

Agenda

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2. **"The science" is not enough**

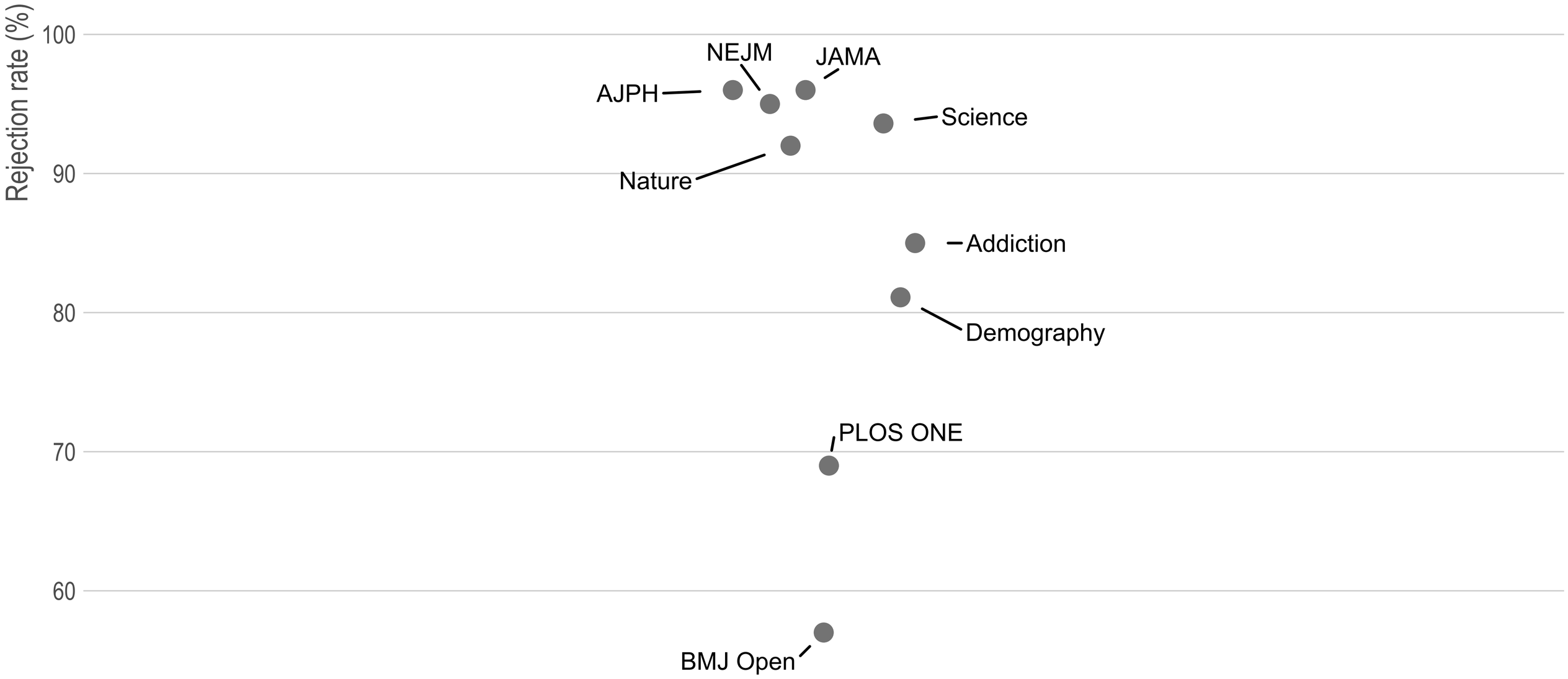
Agenda

1. **Embrace rejection**
2. **"The science" is not enough**
3. **Criticism as collaboration**

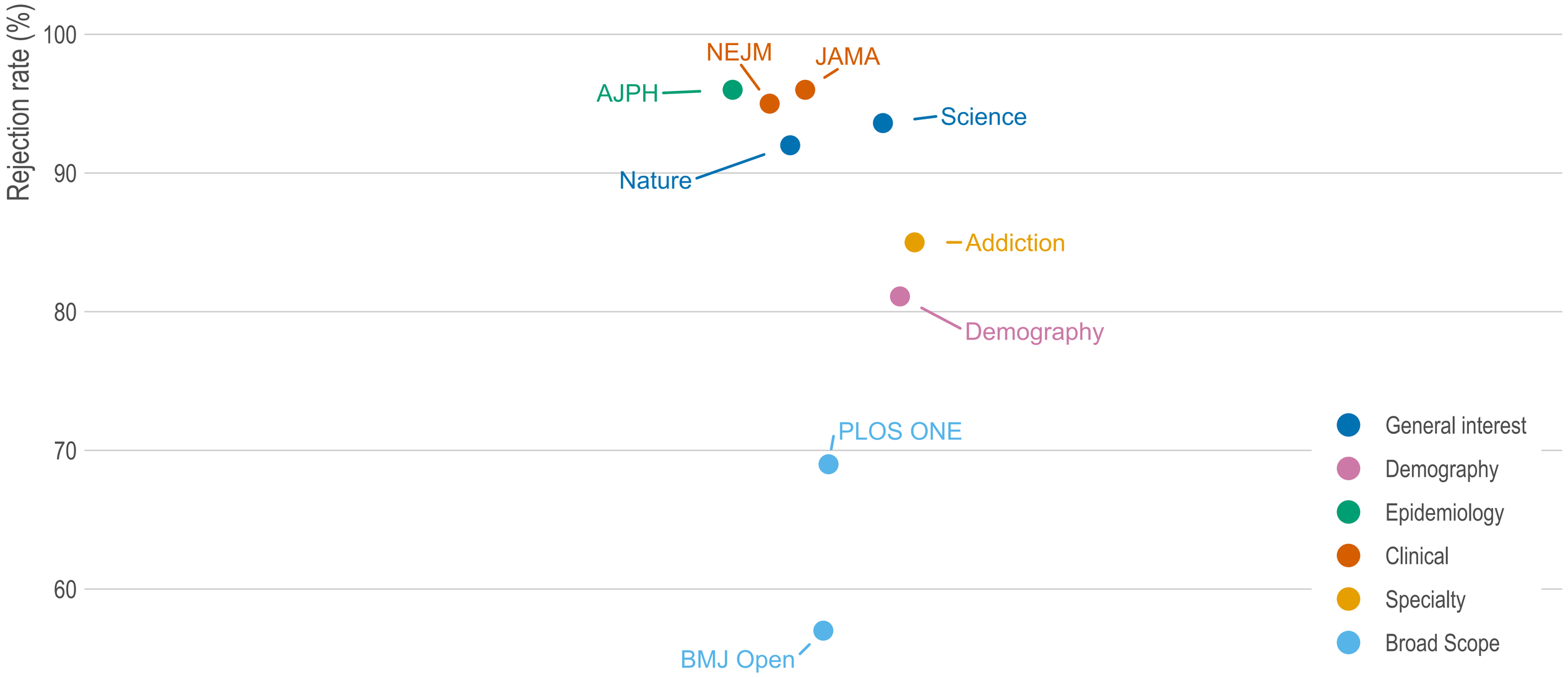
Embrace rejection

(strategically)

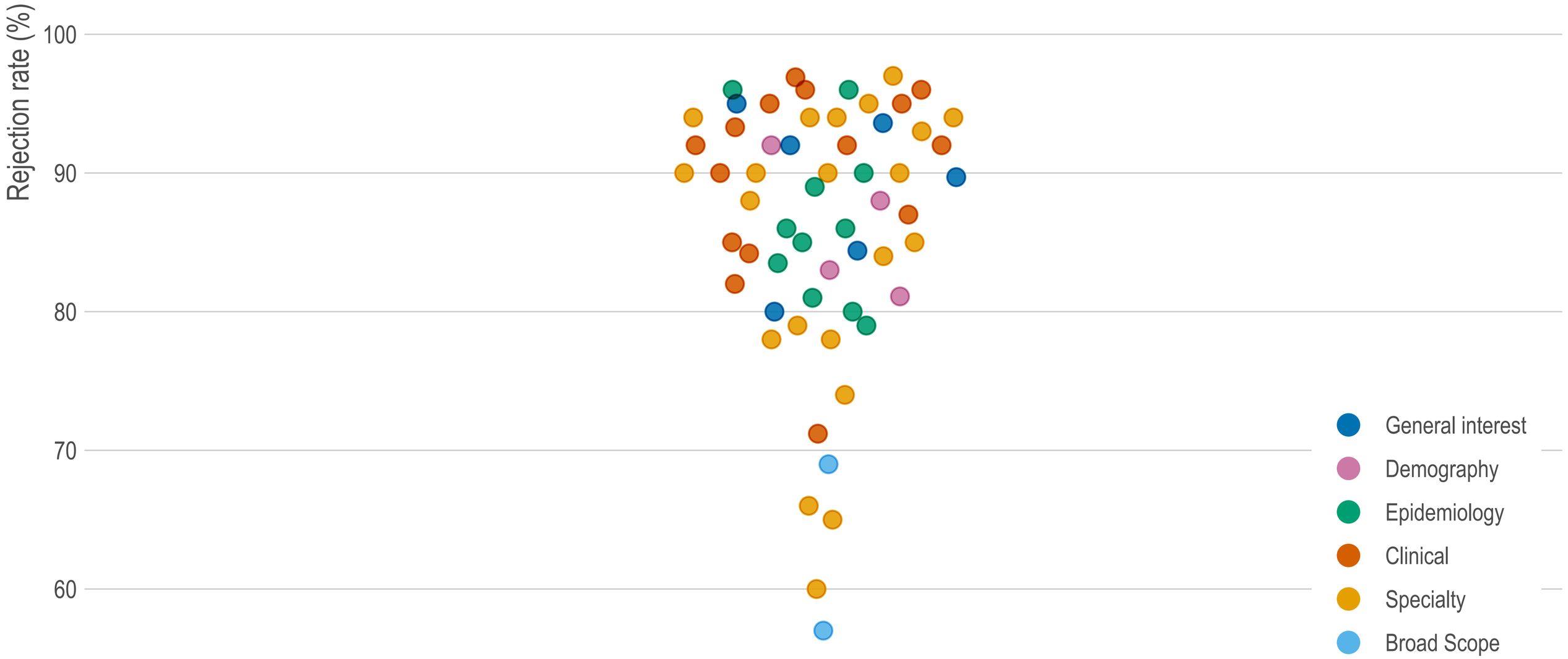
Rejection is the default



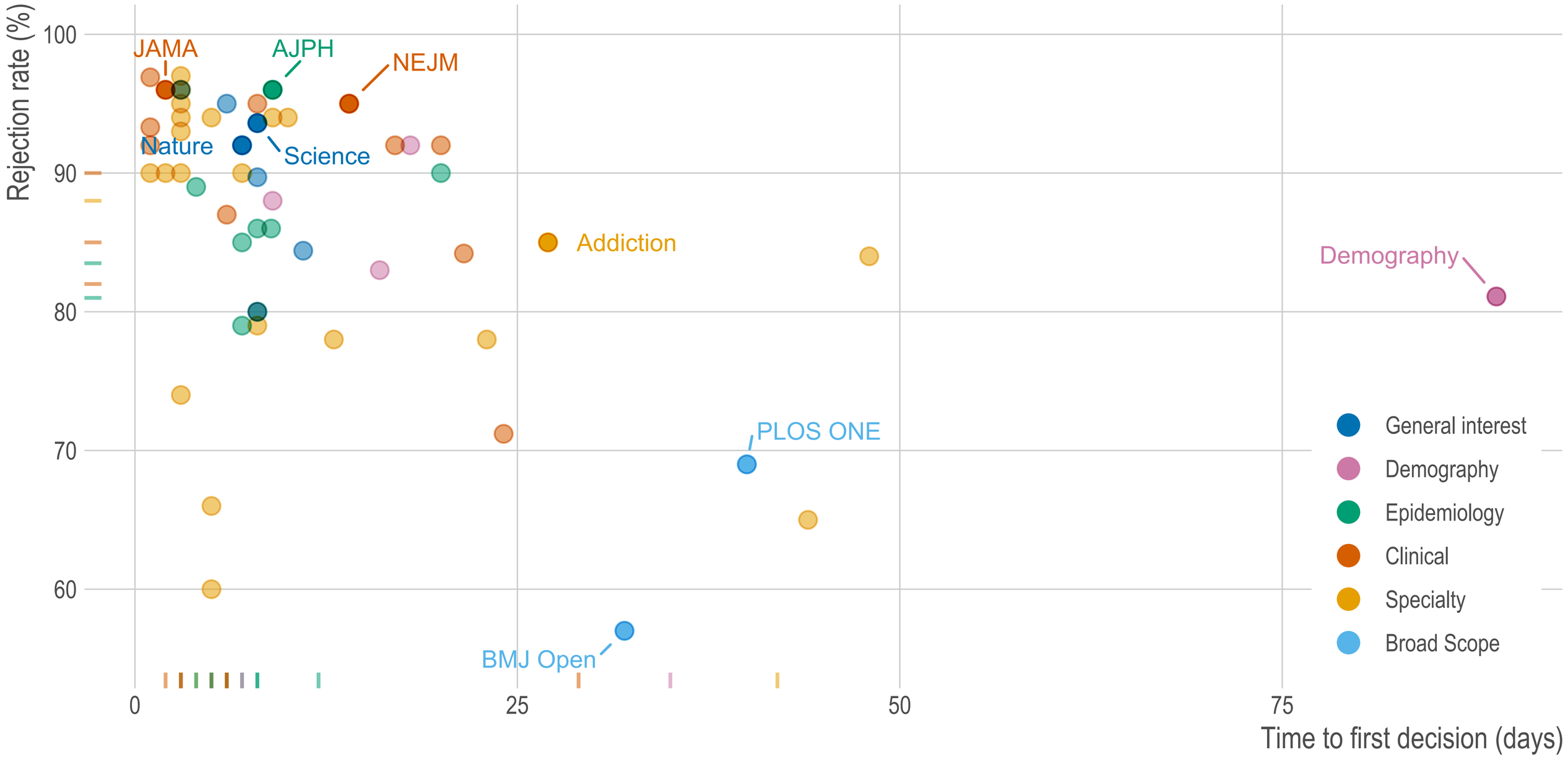
Type of journal matters



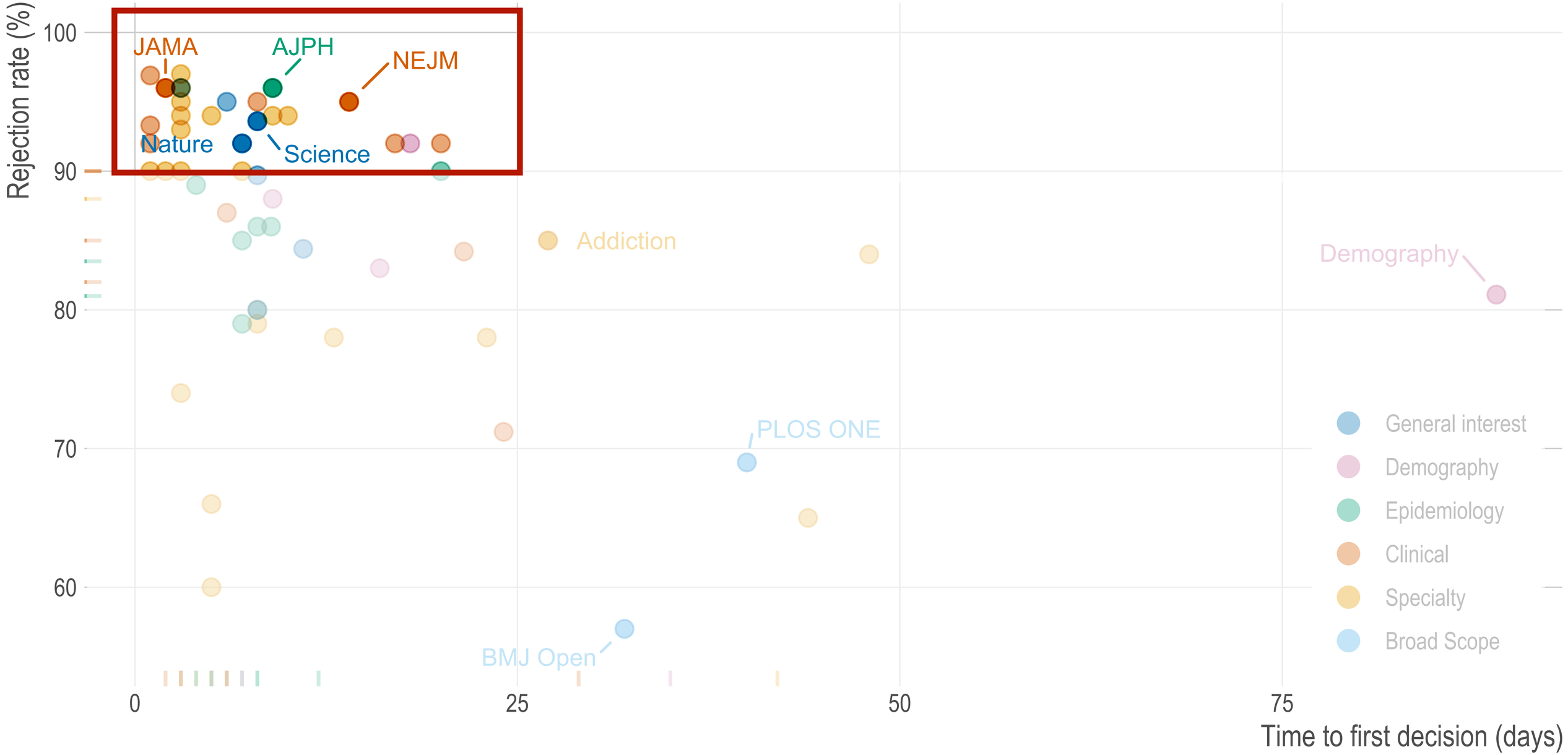
But not that much



Time to first decision has more variation



Desk rejection is usually fast



Peer review is noisy

CONCLUSION

So peer review is a flawed process, full of easily identified defects with little evidence that it works. Nevertheless, it is likely to remain central to science and journals because there is no obvious alternative, and scientists and editors have a continuing belief in peer review. How odd that science should be rooted in belief.

Richard Smith (former editor-in-chief of BMJ)

Peer review: a flawed process at the heart of science and journals.

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Embracing rejection

Aim high

- **Aim high**
  - Low cost: A couple more days to a months- or years-long project
  - High value: Top journals have more reach, infrastructure for media, etc.

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- **If done strategically, an immediate new submission is fast**
  - Minutes to hours not days to weeks
  - Consider journals that have flexible initial submissions:
    format-free (e.g., Nature family journals),
    your paper your way (most Elsevier journals), or
    bioRxiv-to-journal (e.g., Science, some PLOS journals)
    - `https://asntech.github.io/format-free-journals/`

Get rejected fast

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- **Get rejected fast**
  - Have your next journal ready
  - Think of submission *pathways* not *targets*

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- **Get rejected often**
  - Reviewer agreement is barely better than chance[1]
  - A great peer review can change the trajectory of your paper

[1] Kravitz RL, Franks P, Feldman MD, Gerrity M, Byrne C, et al. (2010) Editorial Peer Reviewers' Recommendations at a General Medical Journal: Are They Reliable and Do Editors Care?. PLOS ONE 5(4): e10072. https://doi.org/10.1371/journal.pone.0010072

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"The science" is not enough

"Is this paper likely to make an impact?"

Desk rejection rates

- Nature: ~80%
- JAMA IM: ~80%
- PLOS Medicine: ~80%
- Science / Science Advances: ~85%
- Nature Human Behavior: ~85%
- AJPH: ~90%

Top 10 reasons papers get desk rejected

- Inadequate study design.
- Incomplete methodological details.
- Poor presentation.
- Out of scope of the journal.
- Data availability issues.
- Local interest.
- No novelty.
- Ethical issues.
- Poor English.
- Plagiarism.

Dantas-Torres *Parasites & Vectors* (2022) 15:418
https://doi.org/10.1186/s13071-022-05543-w

Parasites & Vectors

EDITORIAL Open Access

Top 10 reasons your manuscript may be rejected without review

Filipe Dantas-Torres*

Good science is not enough

- **Inadequate study design.**
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Parasites & Vectors

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Your goal:
Get past the editor's desk

Editorial

https://doi.org/10.1038/s41551-022-00957-4

How to make cover letters instructive

Check for updates

Succinctly convey the study's context, emphases, implications and

The title of this Editorial may be read as implying that cover letters to articles submitted to Nature Biomedical Engineering are neither useful nor informative.

Editorial

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How to make cover letters instructive

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Indeed, most aren't.

Editorial

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How to make cover letters instructive

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Cover Letter Basics

The editor is your only audience

- **You are writing for an audience of one.**
  - Editors want to know: Why is this important? Why should I care?

Context, context, context

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  - Editors want to know: Why is this important? Why should I care?
- **Context**: What is the problem, what has been done before, how does this paper fit in relation to work published *in my journal*?

Why should I care?

- **You are writing for an audience of one.**
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- **Context**: What is the problem, what has been done before, how does this paper fit in relation to work published *in my journal*?
- **The advance**: What is new? Be precise — a new data source, a new population, a new method, a better estimate, a new causal identification strategy?

What's the impact?

- **You are writing for an audience of one.**
  - Editors want to know: Why is this important? Why should I care?
- **Context**: What is the problem, what has been done before, how does this paper fit in relation to work published *in my journal*?
- **The advance**: What is new? Be precise — a new data source, a new population, a new method, a better estimate, a new causal identification strategy?
- **(Realistic) Implications and limitations**: What would actually change if we published this paper? Be honest.
  - "The credibility of the inferences improve when they are suitably constrained." (Nat Biomed Eng, 2022)

Cover Letter Template

Hook and importance early

- **Paragraph 1: The hook** and **importance** (2-3 sentences). What problem are you addressing? Why should I care?

Hook and importance early

- **Paragraph 1: The hook** and **importance** (2-3 sentences).
  What problem are you addressing? Why should I care?

  - *As you are aware, the US childhood vaccination schedule is under attack — due to already low vaccination rates, measles outbreaks continue in Texas and New Mexico; states such as Louisiana are ending mass vaccine promotion; and RFK Jr. has indicated that he will reevaluate the ACIP roster. At this time, it is critically important to inform both the American public and policymakers of the potential impact of reducing or removing vaccines.*

Headline findings

- **Paragraph 1: The hook** and **importance** (2-3 sentences). What problem are you addressing? Why should I care?

- **Paragraph 2: The key findings** or advance (3-5 sentences). What you did and what you found. Provide high-level key information (not details).

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  - *In this paper, we ask, "How will declining childhood vaccination affect the risk of outbreaks for previously eliminated infectious diseases in the US (measles, rubella, diphtheria, poliomyelitis)?" We used stochastic dynamical models to estimate the impact of vaccine reductions (at different levels) over 25 years. We found: (1) Even under current vaccination rates, measles is likely to become endemic... (2) A decline of just 5% from current state-level vaccination rates would result in...*

Show me the fit

- **Paragraph 1: The hook** and **importance** (2-3 sentences). What problem are you addressing? Why should I care?

- **Paragraph 2: The key findings** or advance (3-5 sentences). What you did and what you found. Provide high-level key information (not details).

- **Paragraph 3: Why this journal** (3-4 sentences)? Show me we are a good fit, and that our readers would care about this topic.

  - *We believe this paper would be of particular interest to the readership of Journal X, which has recently published papers using similar methods (e.g., doi: 10.001.JOURNALX.1) and argued for the importance of vaccine infrastructure both in editorials (10.001.JOURNALX.2) and original research articles (10.001.JOURNALX.3).*

Logistics

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- **Paragraph 4: Logistics** (2-3 sentences). Show me you read the Instructions to Authors and know the journal requirements.

  - For example: Work is original and not under consideration elsewhere. Has it been published or preprinted elsewhere? Note potential conflicts or discussions with editors, etc.

Cover letter template

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- **Paragraph 4: Logistics** (2-3 sentences). Show me you read the Instructions to Authors and know the journal requirements.
- **Keep it short! Don't include**: Detailed methods, same information as the abstract, potential reviewers or excluded reviewers (only if no other place in the portal), etc.

"The science" is not enough

- **Write your cover letter <u>specifically</u> for the editor.**
  - This means tailoring for every journal!
- **Use the template** (or something very close).
  - Tell the editor early on why they should care.
  - Be specific about your advance.
  - Be transparent about implications and limitations.
  - Cite papers from the same journal.
- **One page.** No jargon.
- (Titles and abstracts are the first things the editor will see. <u>Make them strong</u>.)

Criticism as collaboration

Initial submission
Technical review
Senior editor (SE) triage
Associate editor (AE) review
Peer reviewers
AE synthesizes
SE makes decision

R&Rs

- You beat the odds!

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You could:

- Decline the R&R and go elsewhere
- Appeal the decision

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You could:

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- Appeal the decision

But you shouldn't!

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- Evidence suggests peer review improves paper quality[1] and papers that go through resubmission get more citations[2].

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R&Rs

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- Criticism should be viewed as a collaboration — not an thesis defense.
- Evidence suggests peer review improves paper quality[1] and papers that go through resubmission get more citations[2].
- At this point, **the response letter is by far the most important document**.

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Your goal:
Avoid going back out for review

PLOS | COMPUTATIONAL BIOLOGY

EDITORIAL

Ten simple rules for writing a response to reviewers

William Stafford Noble*

Departments of Genome Sciences and Computer Science and Engineering, University of Washington, Seattle, Washington, United States of America

* william-noble@uw.edu

https://doi.org/10.1371/journal.pcbi.1005730

Ten Simple Rules

1. Provide an overview, then quote full set of reviews.
2. Be polite and respectful of all reviewers.
3. Accept the blame.
4. Make the response self-contained.
5. Respond to every point raised by the reviewer.
6. Use typography.
7. Begin each response with a direct answer.
8. Do what the reviewer asks.
9. Be clear about what changed relative to the previous version.
10. Write the response twice.

Noble WS (2017) Ten simple rules for writing a response to reviewers. PLoS Comput Biol 13(10): e1005730.

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11. **Get it done now.**

Noble WS (2017) Ten simple rules for writing a response to reviewers. PLoS Comput Biol 13(10): e1005730.

Make the response self-contained

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Literally copying and pasting

- Quote the comment verbatim.

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Directly address the critique

- Quote the comment verbatim.
- Underneath address the critique directly (Yes/No) and then elaborate.

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Show me the response

- Quote the comment verbatim.
- Underneath address the critique directly (Yes/No) and then elaborate.
- Provide the original text, provide the changed text with <u>emphasis on text that addresses the point</u> along with the page and line number of the change.

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Support your claims

- Quote the comment verbatim.
- Underneath address the critique directly (Yes/No) and then elaborate.
- Provide the original text, provide the changed text with emphasis on text that addresses the point along with the page and line number of the change.
- Copy the table, figure, full reference, etc. that supports your claim.

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- Provide the original text, provide the changed text with emphasis on text that addresses the point along with the page and line number of the change.
- Copy the table, figure, full reference, etc. that supports your claim.
- **The associate editor should be able to assess if you adequately addressed the comment without ever opening the manuscript.**

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Respond to every point

Respond to every point

- If a paper goes back out for review, by far the most frequent critique in the second round of review is that the authors didn't respond to a critique in the first round. **Avoid this**.

- Number the reviewer comments (e.g., R1-1, R2-10, R3-3, E-1), **splitting out double-barreled critiques**.

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- Number the reviewer comments (e.g., R1-1, R2-10, R3-3, E-1), **splitting out double-barreled critiques**.

- Respond to <u>all</u> points even if you disagree, but **never** just ignore a point.

Do what the reviewer asks

- The second most frequent critique on a second round revision is that the authors responded to the critique by not doing anything. **Don't do this.**

Do what the reviewer asks

- The second most frequent critique on a second round revision is that the authors responded to the critique by not doing anything. **Don't do this.**

- You should try to do what the authors asked, even if it is just a supplemental analysis that never makes it into the publication or appendix. If you think it doesn't matter, <u>show</u> us that it doesn't matter.

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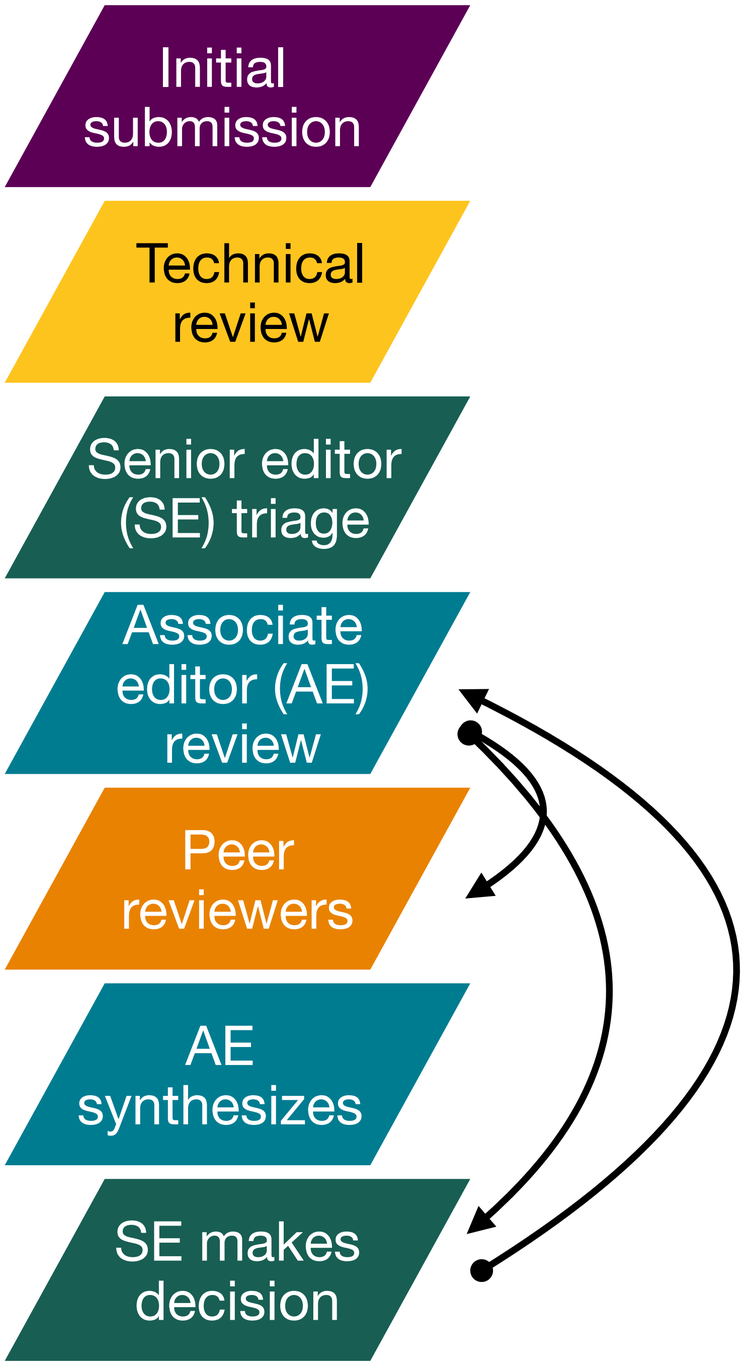
- You should try to do what the reviewers asked, even if it is just a supplemental analysis that never makes it into the publication or appendix. If you think it doesn't matter, <u>show</u> us that it doesn't matter.
- If the reviewer is asking for too much, do **some** of it. Don't just say everything is "out of scope".

Get it done now

Criticism as collaboration

- **R&Rs are invitations** to get your paper published
- Response letters are **high leverage documents** — spend time on them!
- **Avoid going back out for review again**
  - Respond quickly
  - Respond thoroughly
  - Respond respectfully
  - Make the editor's job (and potential reviewers' jobs) as easy as possible (use typography, responses are self-contained, etc.)



Embrace rejection

"The science" is not enough

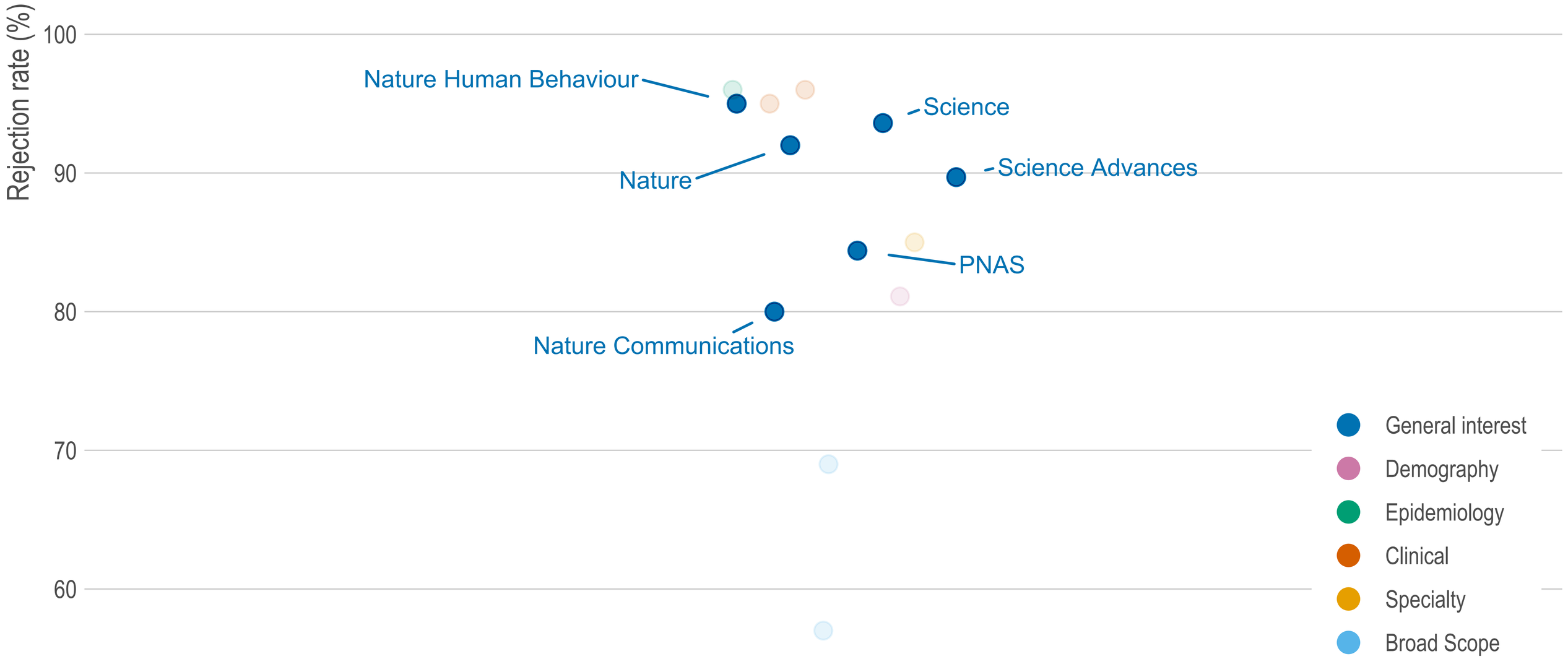
Criticism as collaboration

Thank you

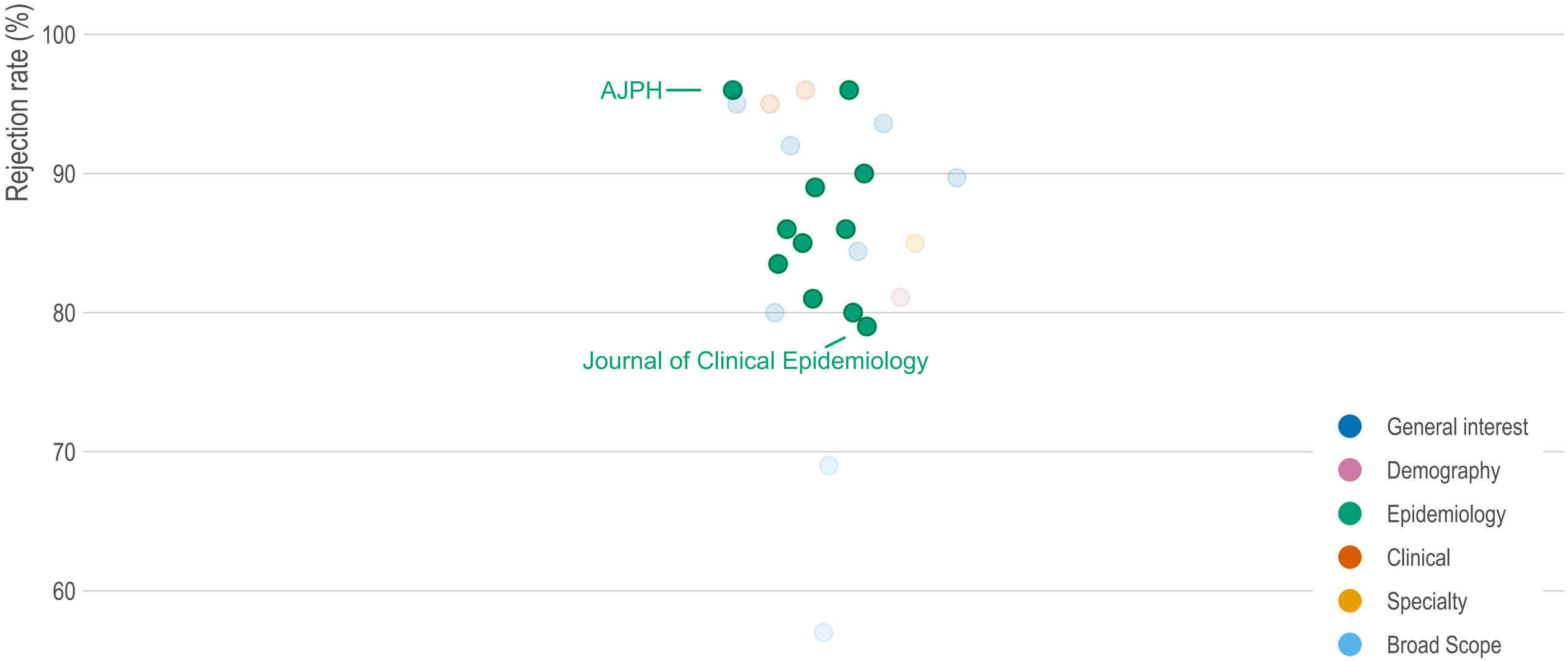
Slides, data, code, and more resources at:
`https://github.com/mkiang/paper_to_publication`

Extra slides

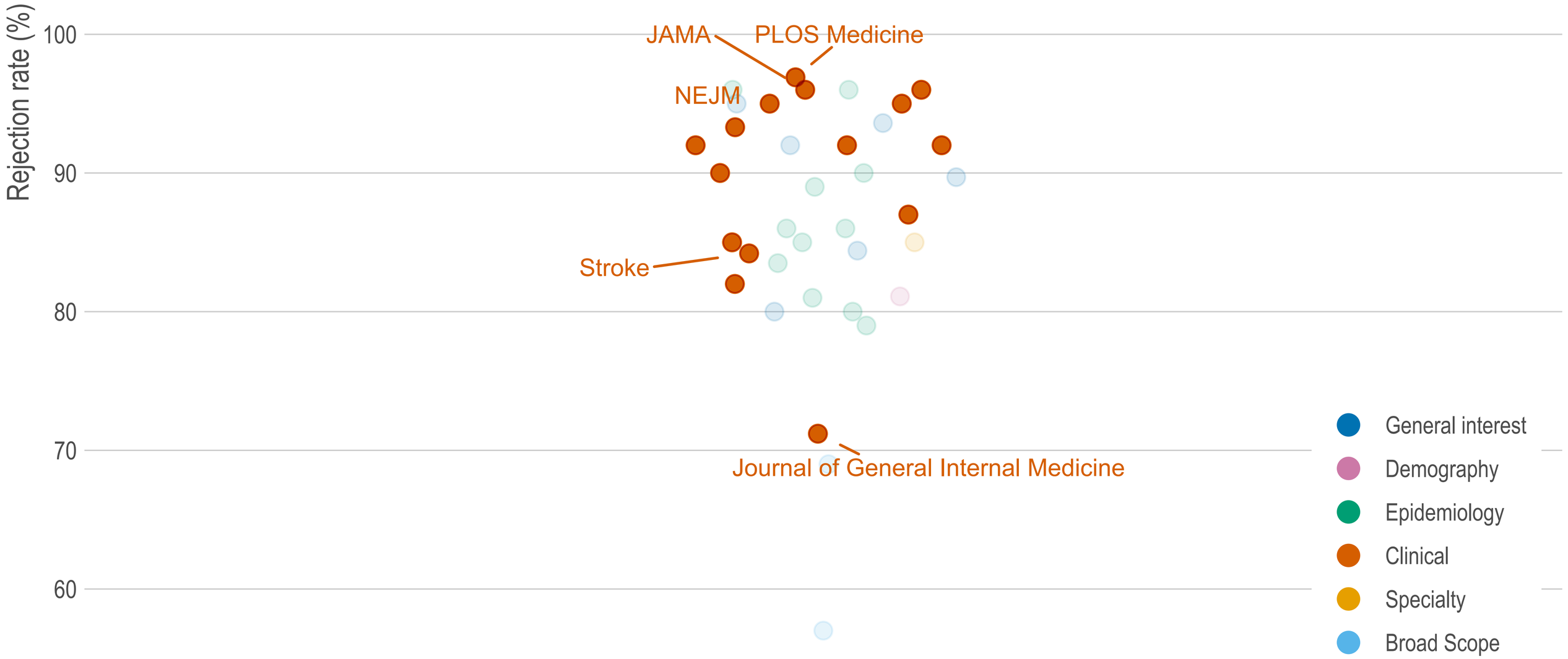
General interest journals



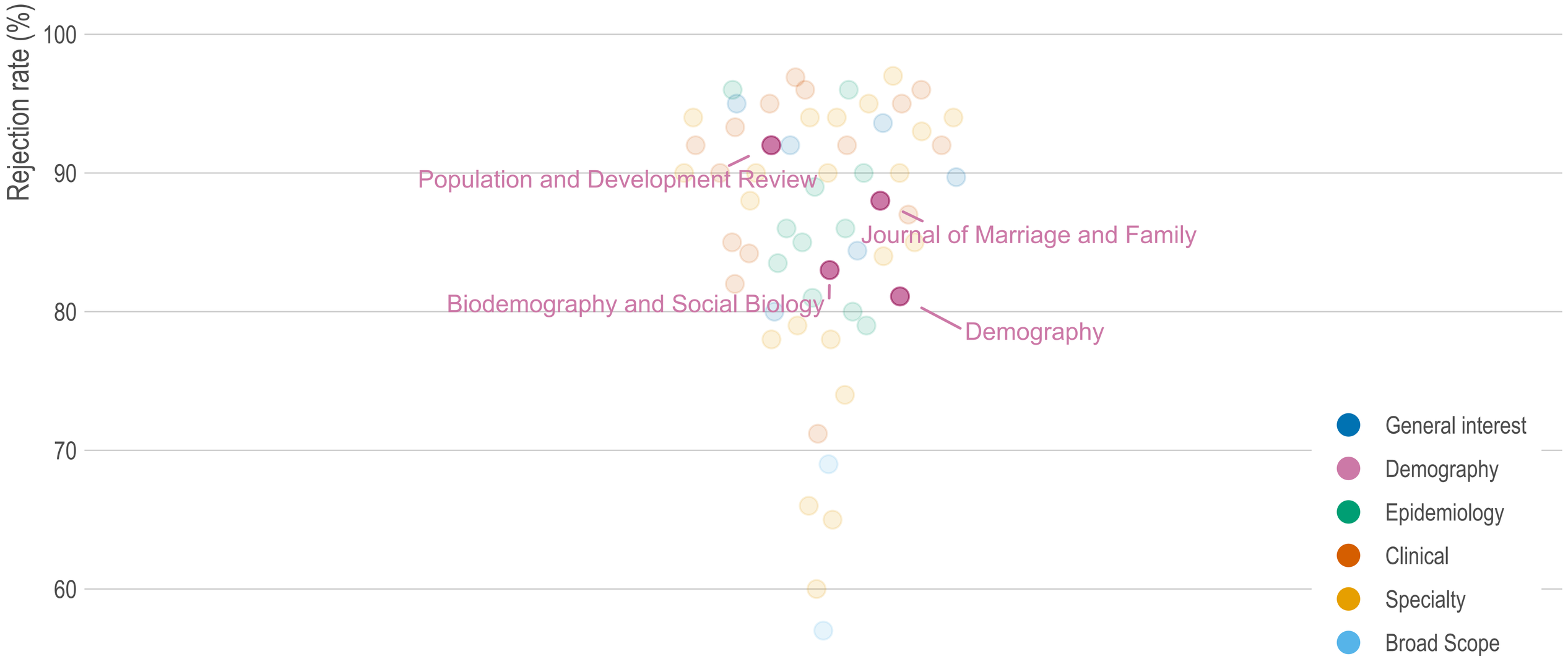
Epidemiology journals



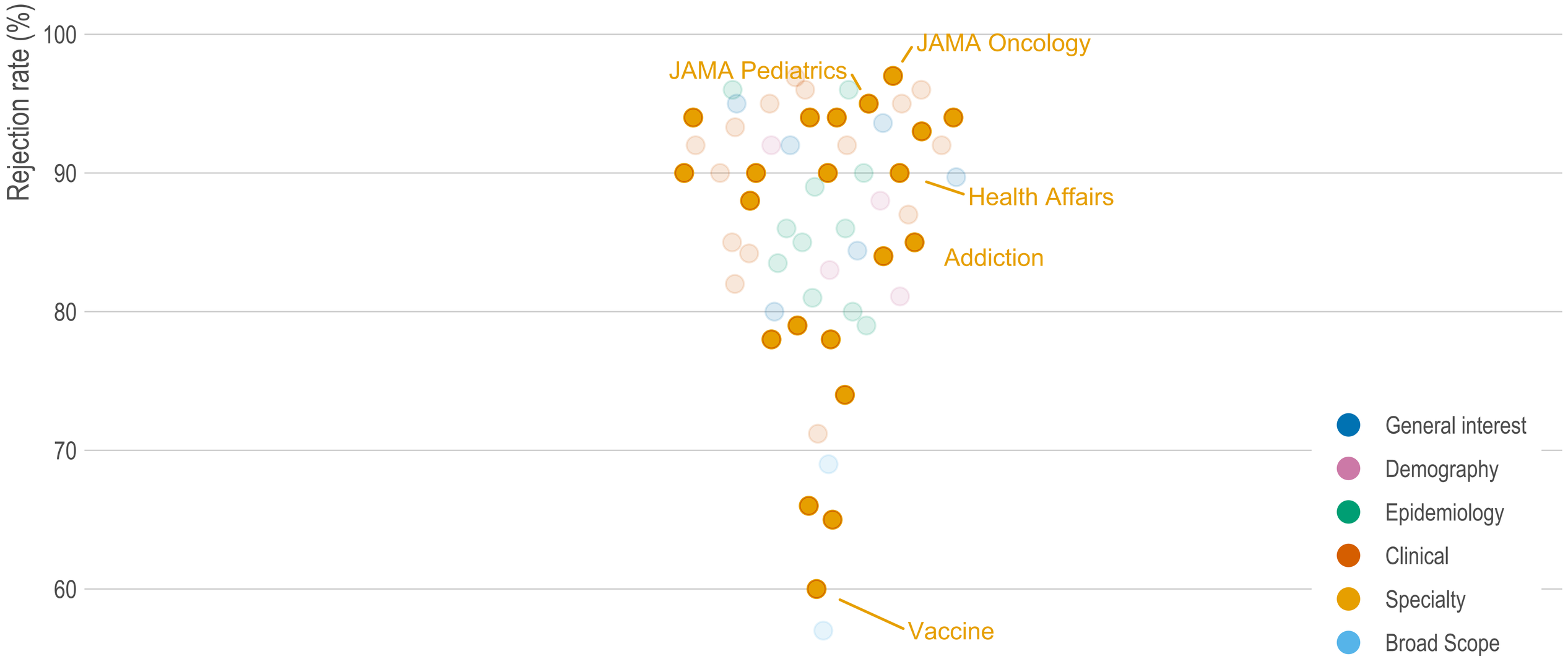
Clinical journals



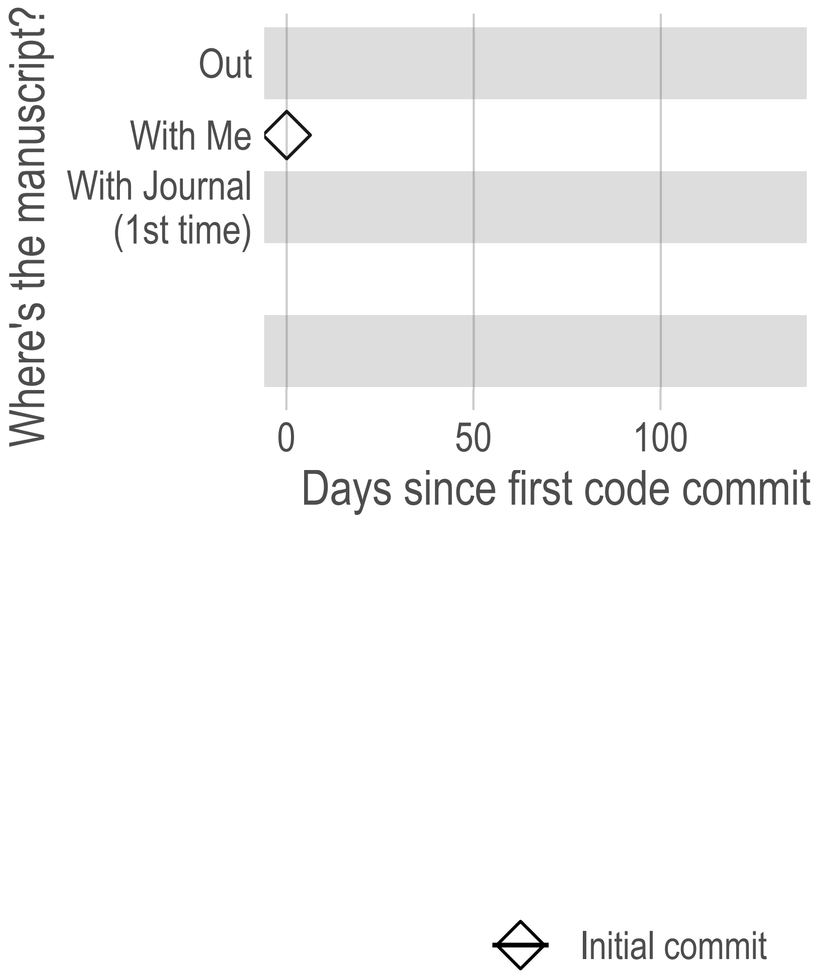
Demography journals



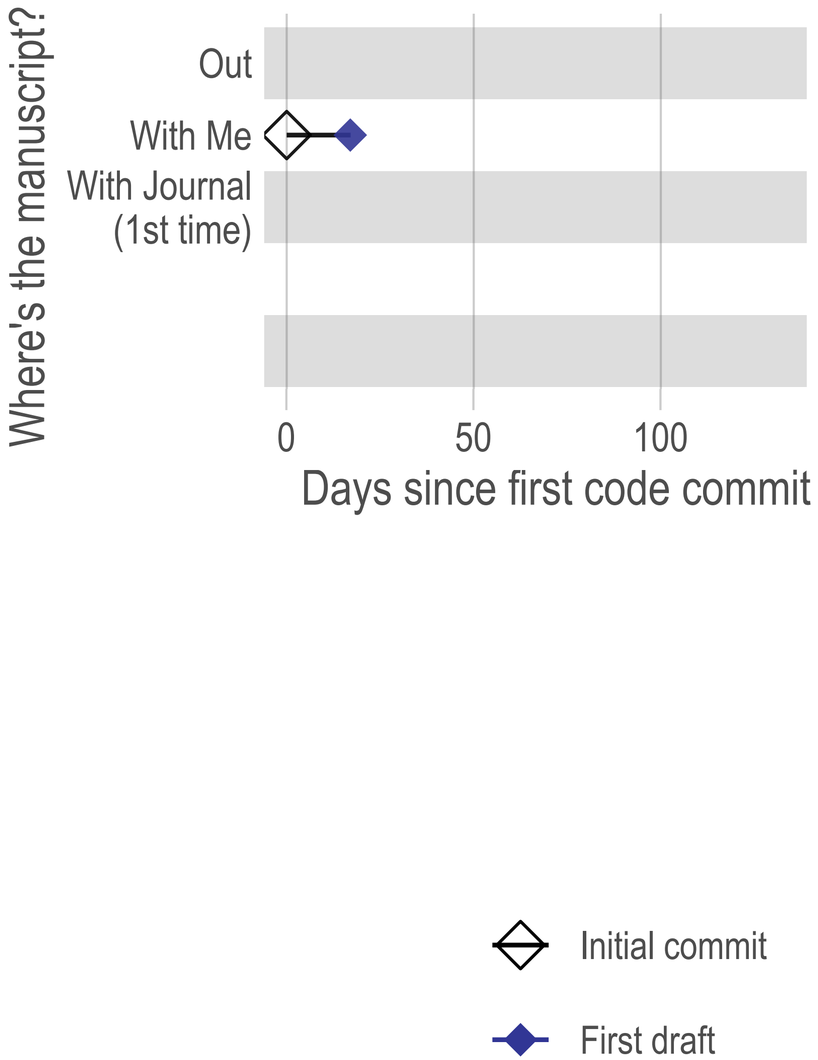
Specialty journals



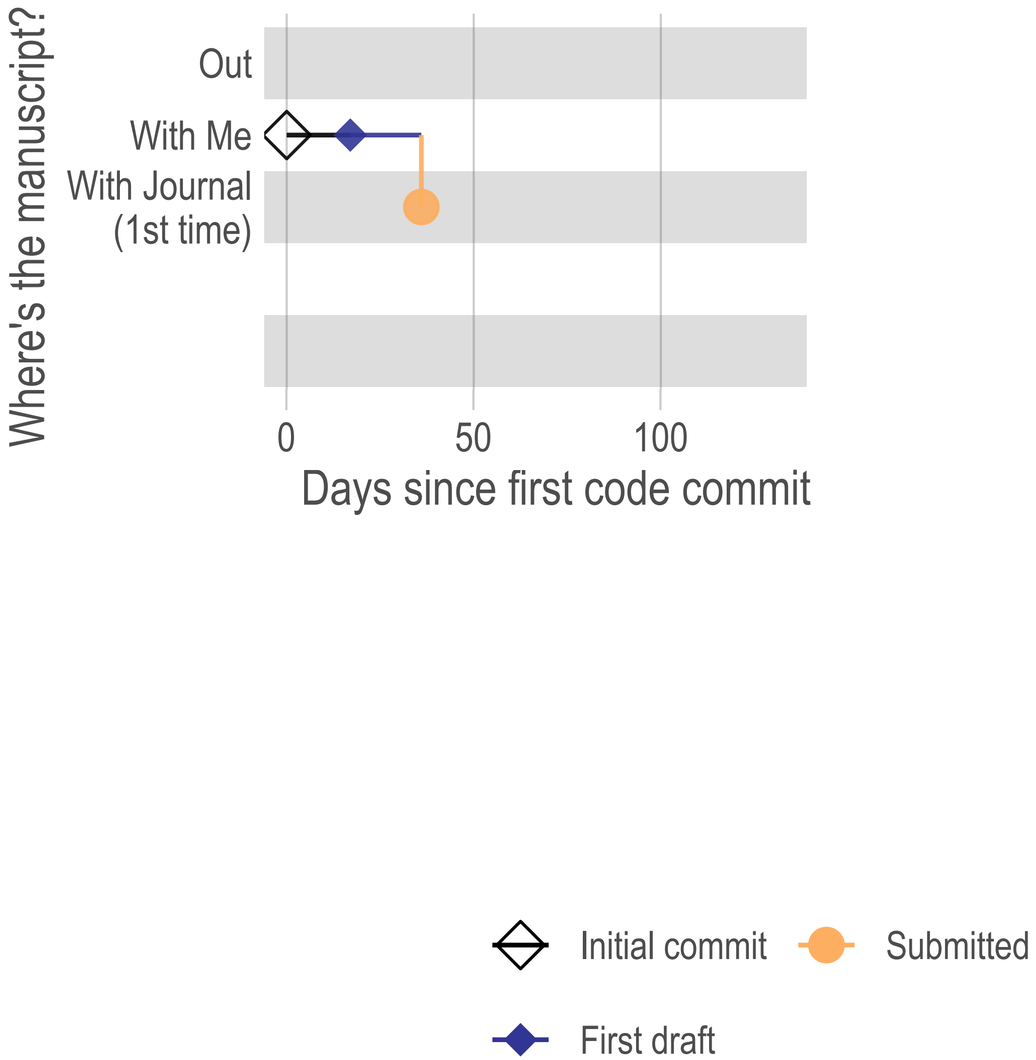
Full analytic code completed



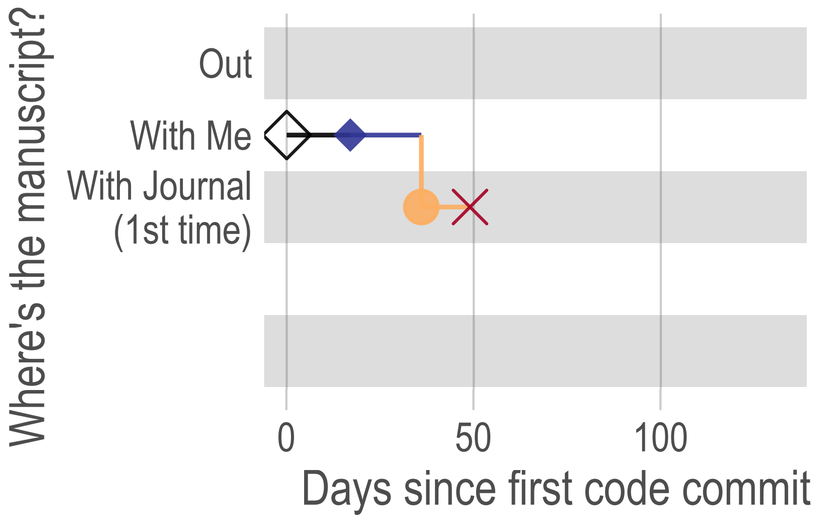
First draft written 14 days later



Paper submitted 22 days later

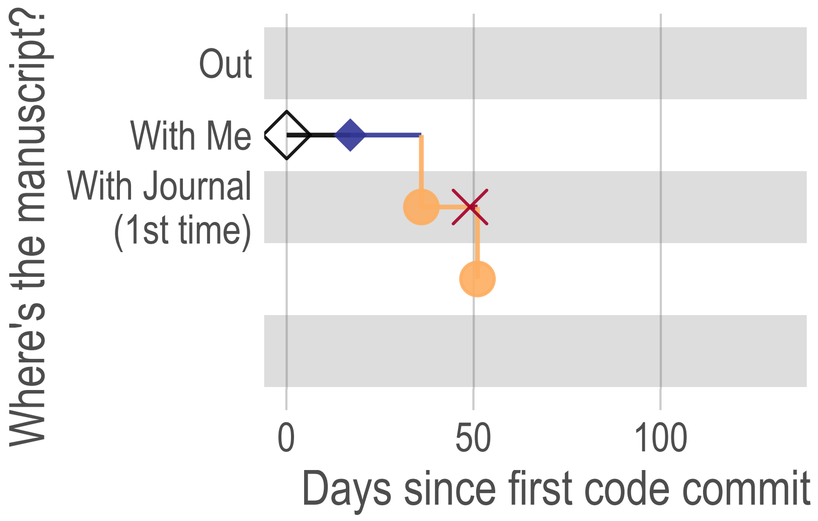


Desk rejected 13 days later



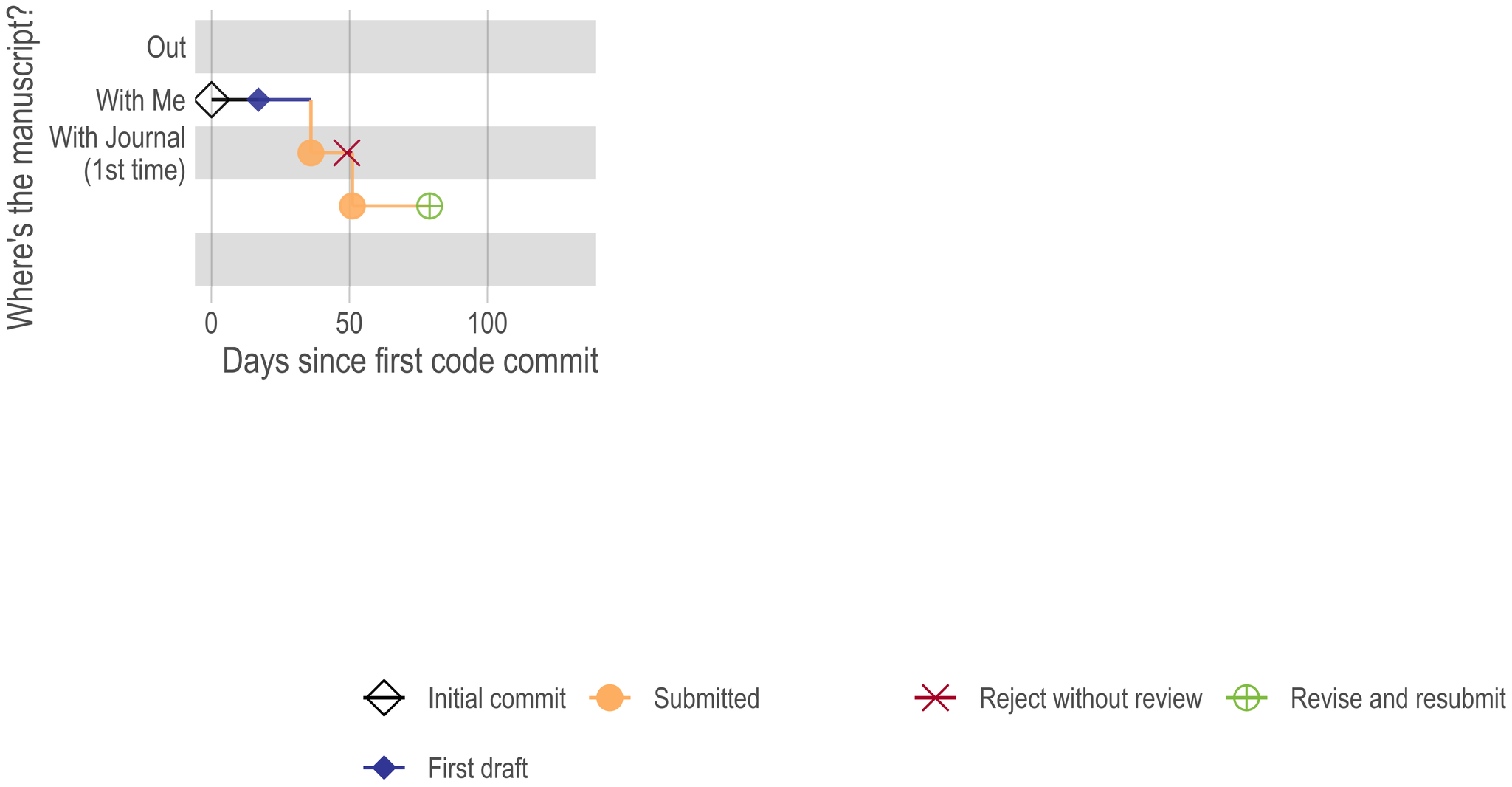
Initial commit
Submitted
Reject without review
First draft

Resubmitted 2 days later

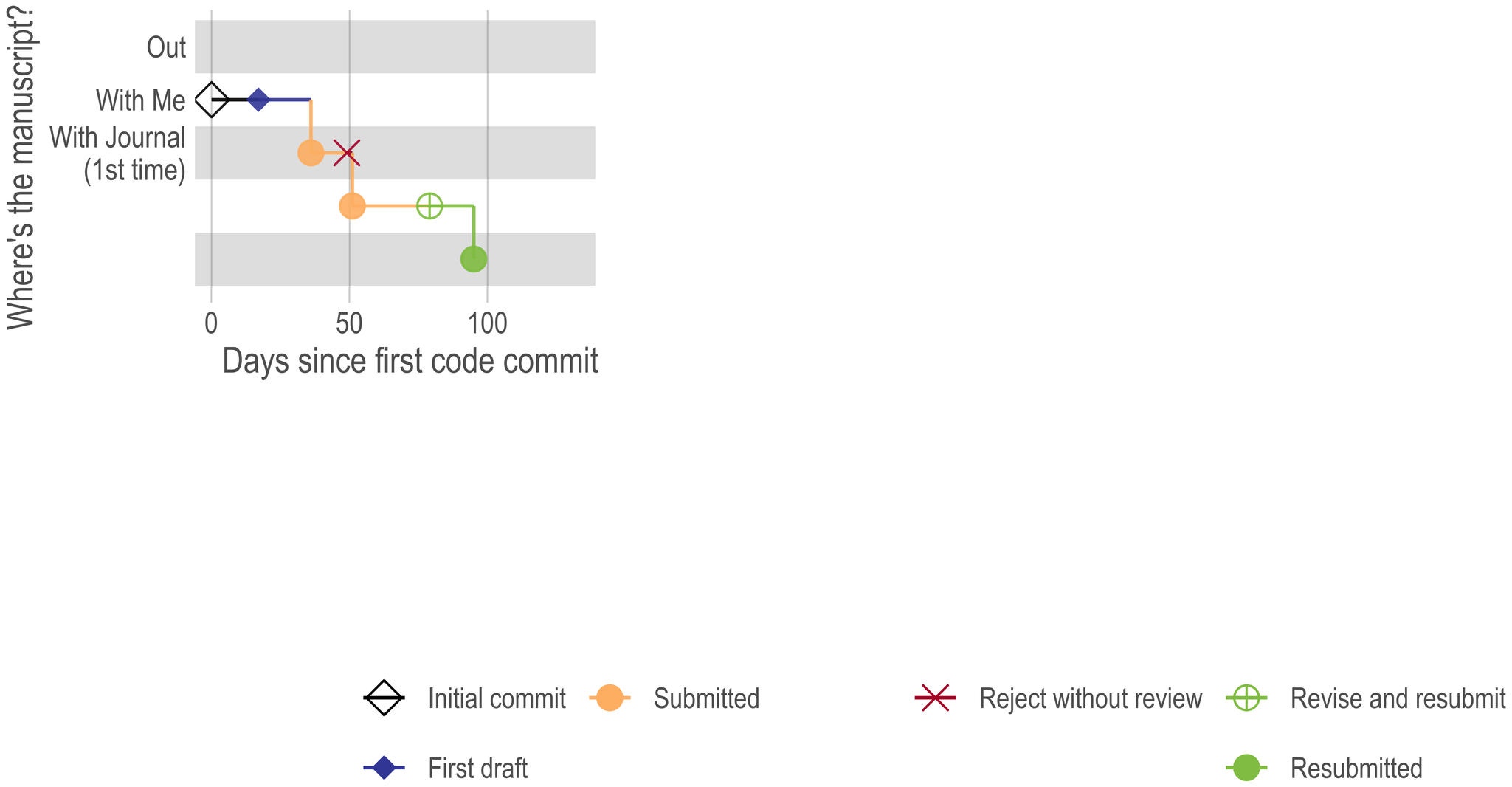


Initial commit
Submitted
Reject without review
First draft

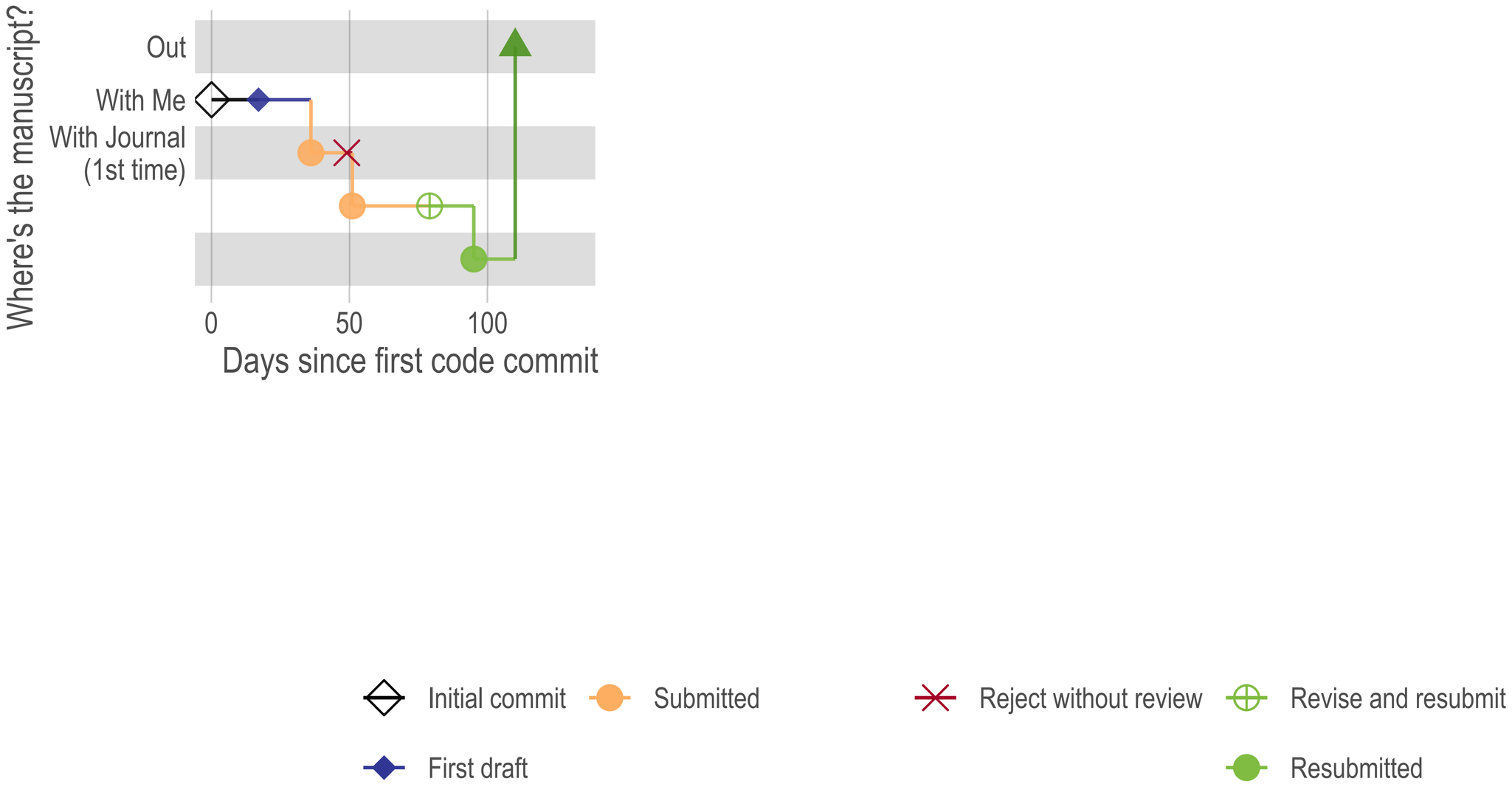
Revise and resubmit decision 27 days later



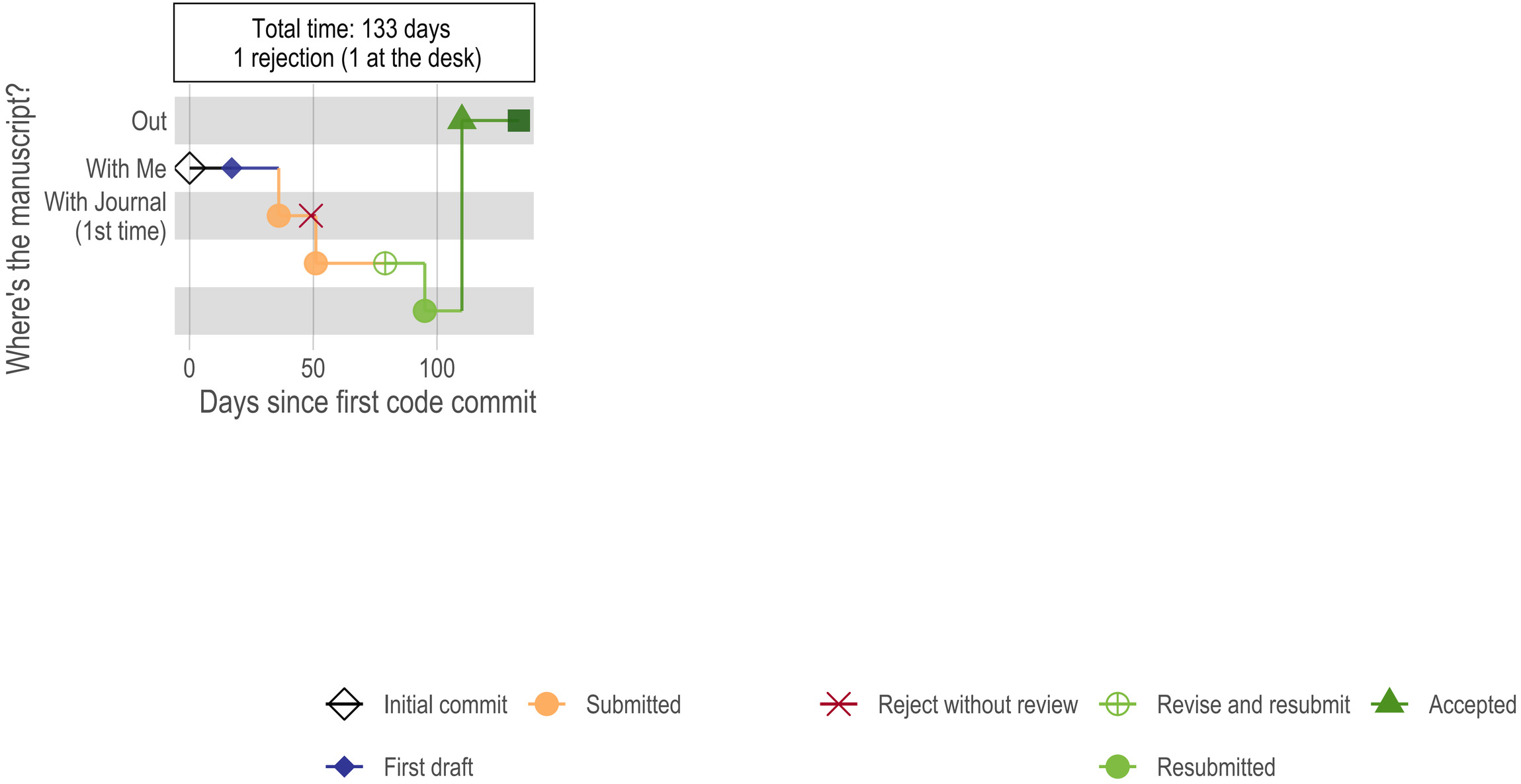
Revision submitted 16 days later



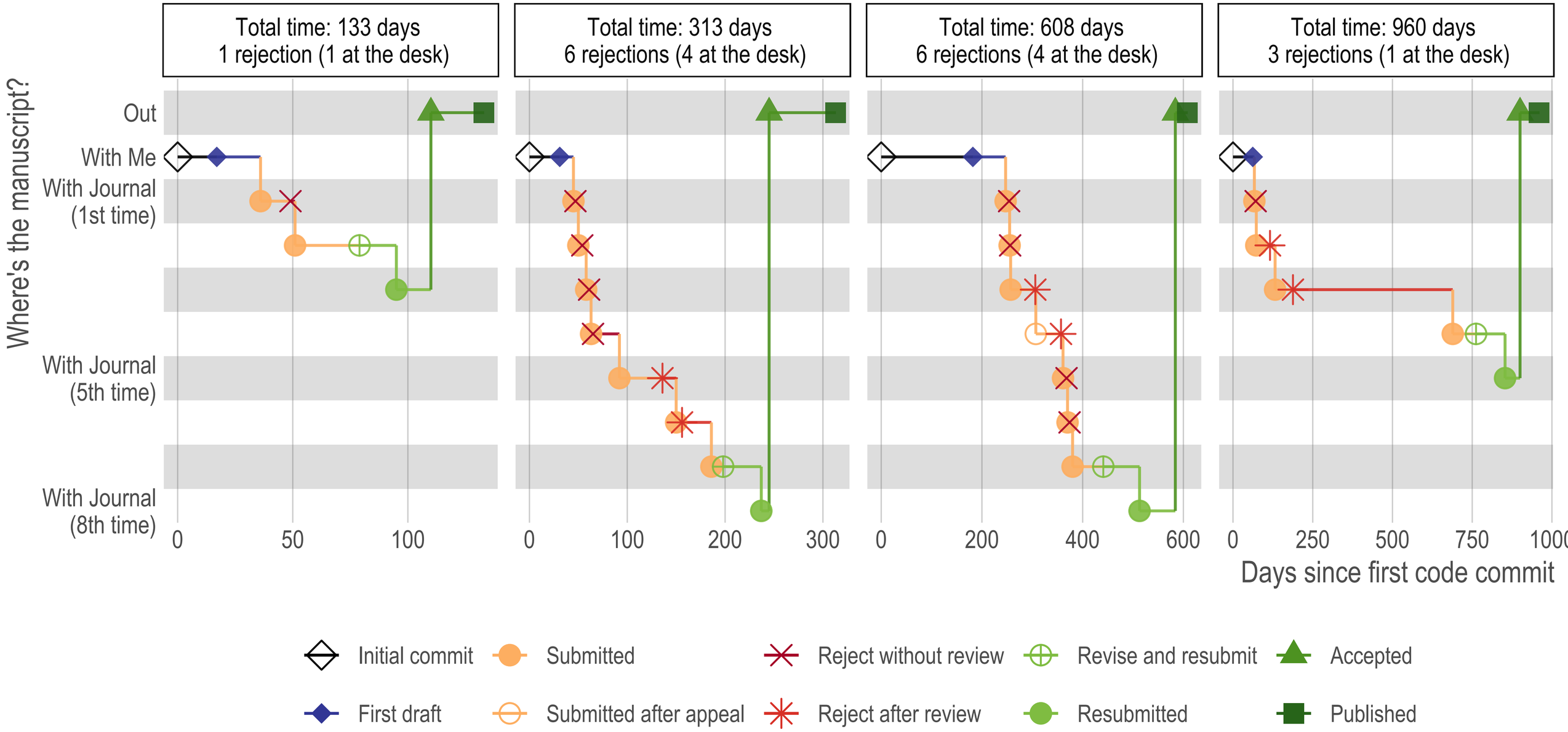
Accepted 15 days later



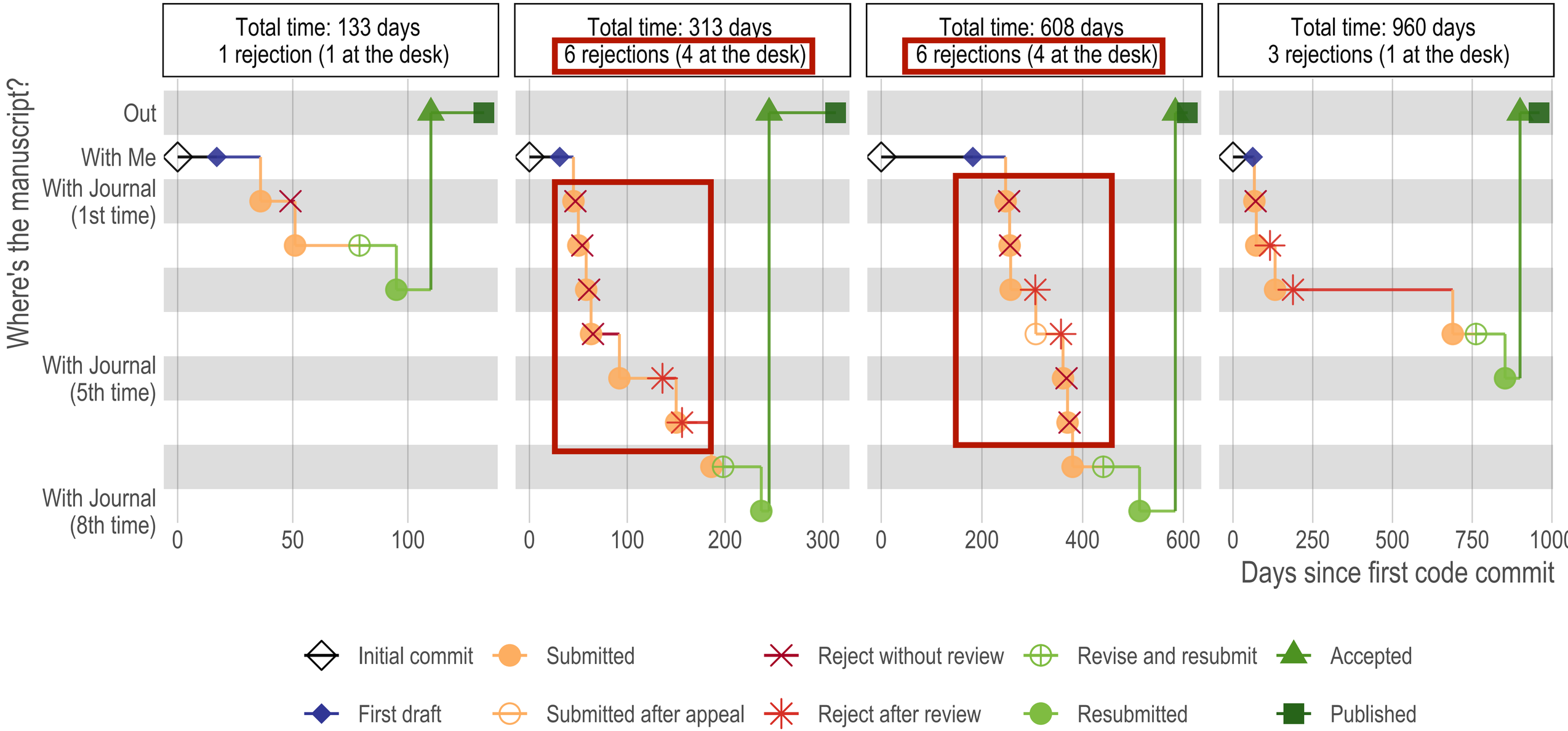
Published 23 days later



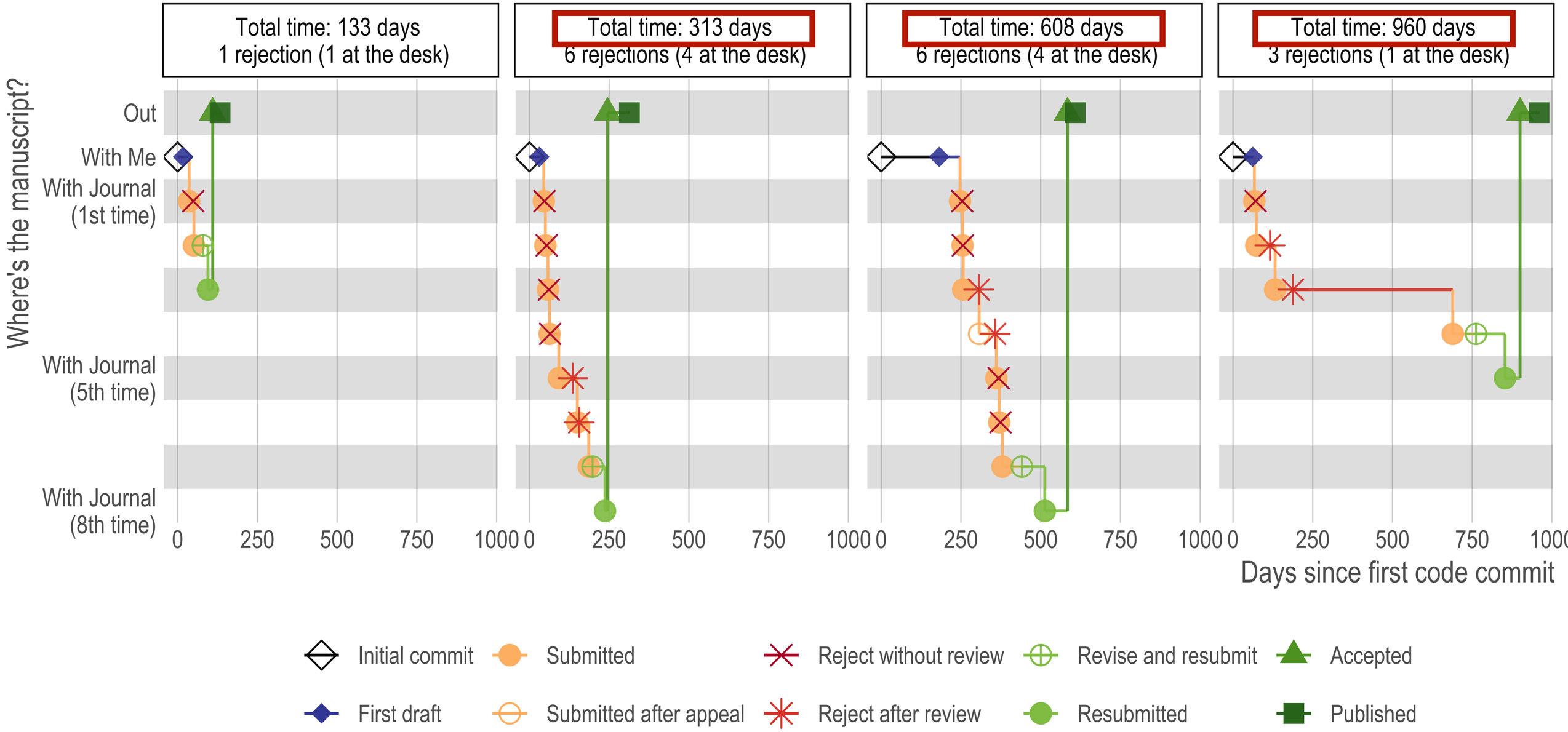
Not my typical timeline



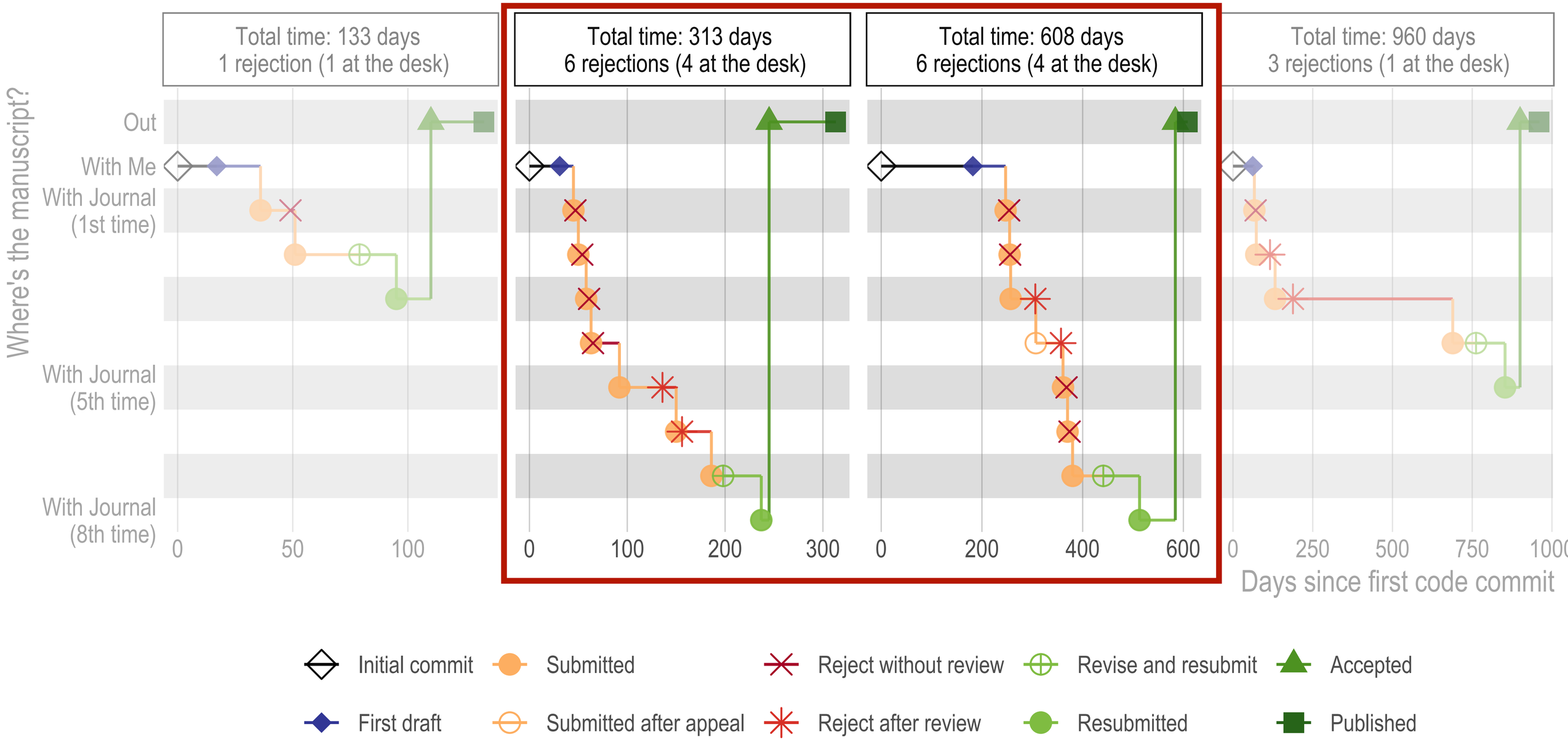
My papers get rejected a lot



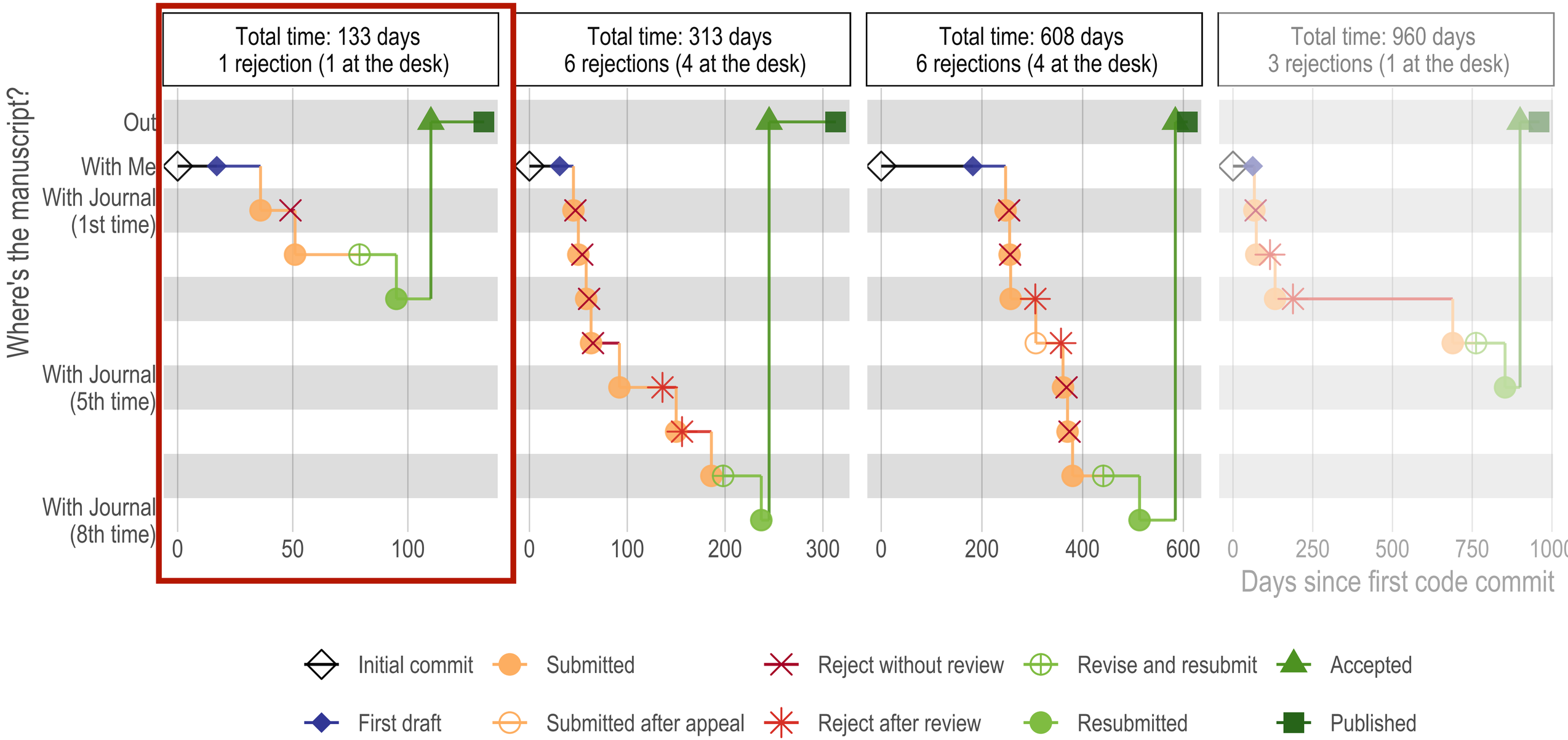
And take a long time



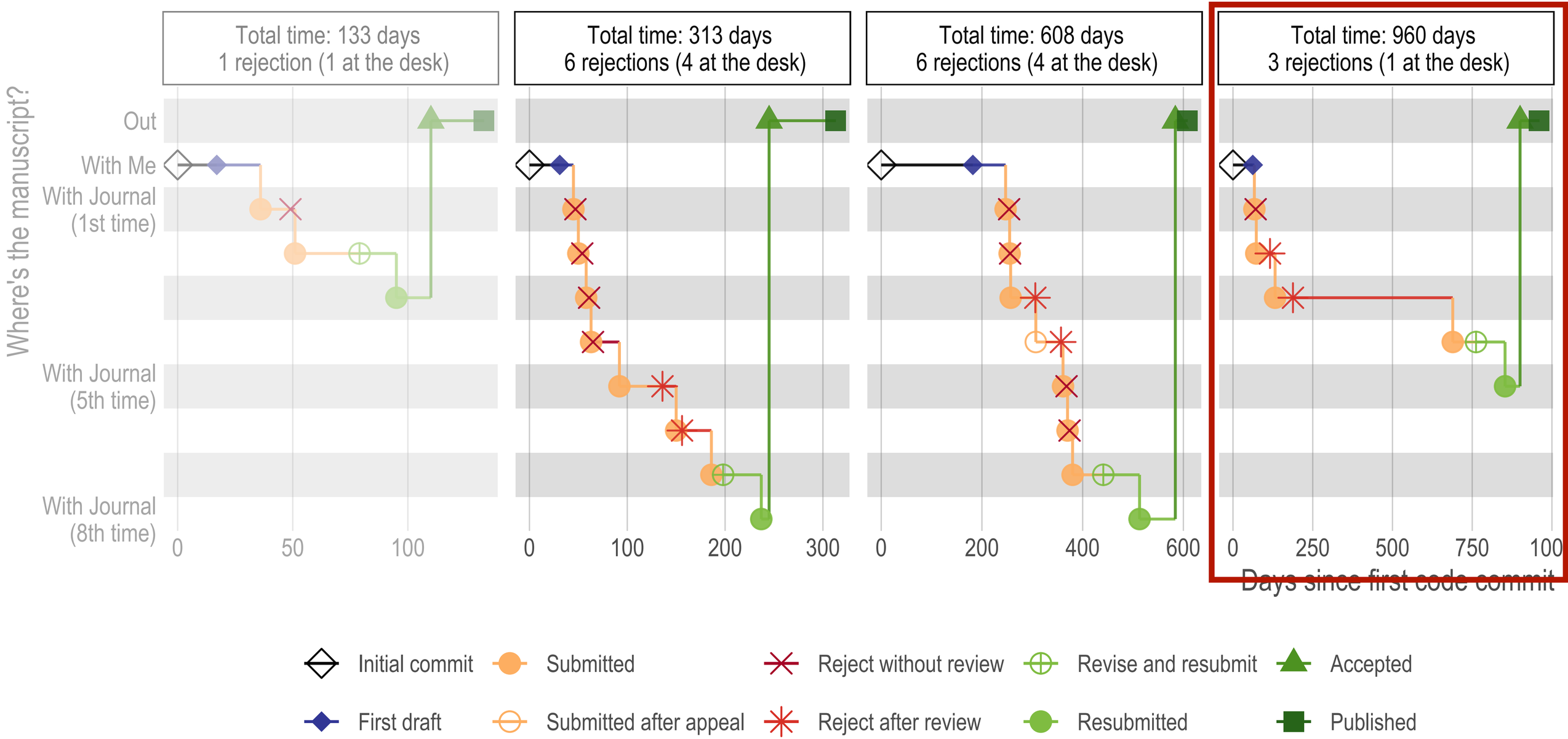
This is actually what I am aiming for



Sometimes it works out



Peer review can greatly improve your paper



Cover Letter Tips

- Template won't work for all papers but should work for most.
- One page max.
- Don't just repeat the abstract.
- Always use references from the target journal. If you can't find related references **it's probably the wrong journal.**

Title

- **Use a strong title.** (Even if it doesn't adhere to journal standards.)
  - First thing an editor reads.
  - Weak title: Could be describing a thousand papers, unclear scope, unclear time period.
  - Strong title: Editor immediately knows the finding, the scope, and time period.

Weak	Strong
An analysis of opioid-related mortality trends in the United States	Opioid mortality declined in 38 states after naloxone access laws, 2010-2022

The Abstract's "Problem Statement"

- **So what?**
  - Second thing the editor reads.
  - Should answer two questions:
    - Why should I care about *this*?
    - Why should I care about this *right now*?
- Policy relevance, clinical relevance, or filling a specific evidence gap are the clearest signals.

Weak

Opioid use disorder is a major public health crisis in the United States.

Strong

Whether state naloxone access laws reduce opioid mortality remains unclear despite adoption by X states.